1) a)

$\frac{\mathrm{d}y}{\mathrm{d}t} = y$

Use Separation of Variables

$\Rightarrow \int \frac{1}{y}\mathrm{d}y = \int 1\,\mathrm{d}t$

$\Rightarrow \ln|y| = t + c$

$\Rightarrow y = A\mathrm{e}^{t}$

1) b)

$\frac{\mathrm{d}x}{\mathrm{d}t} = x + A\mathrm{e}^{t}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} - x = A\mathrm{e}^{t}$

Integrating Factor

$= \mathrm{e}^{\int -1\mathrm{d}t}$

$= \mathrm{e}^{-t}$

Let Integrating Factor $= \mathrm{e}^{-t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} - x = A\mathrm{e}^{t}$

$\Rightarrow \mathrm{e}^{-t}\frac{\mathrm{d}x}{\mathrm{d}t} - x\mathrm{e}^{-t} = A\mathrm{e}^{t} \times \mathrm{e}^{-t}$

$\Rightarrow \mathrm{e}^{-t}\frac{\mathrm{d}x}{\mathrm{d}t} - x\mathrm{e}^{-t} = A$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(x\mathrm{e}^{-t}) = A$

$\Rightarrow x\mathrm{e}^{-t} = At + c$

$\Rightarrow x = (At + c)\mathrm{e}^{t}$

2) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x$

Use Separation of Variables

$\Rightarrow \int \frac{1}{x}\mathrm{d}x = \int 2\,\mathrm{d}t$

$\Rightarrow \ln|x| = 2t + c$

$\Rightarrow x = A\mathrm{e}^{2t}$

2) b)

$\frac{dy}{dt} = Ae^{2t} + 2y$

$\Rightarrow \frac{dy}{dt} - 2y = Ae^{2t}$

Integrating Factor

$= e^{\int -2dt}$

$= e^{-2t}$

Let Integrating Factor $= e^{-2t}$

$\frac{dy}{dt} - 2y = Ae^{2t}$

$\Rightarrow e^{-2t}\frac{dy}{dt} - 2ye^{-2t} = Ae^{2t} \times e^{-2t}$

$\Rightarrow e^{-2t}\frac{dy}{dt} - 2ye^{-2t} = A$

$\Rightarrow \frac{d}{dt}(ye^{-2t}) = A$

$\Rightarrow ye^{-2t} = At + c$

$\Rightarrow y = (At + c)e^{2t}$

3) a)

$\frac{dx}{dt} = x + y$

$\Rightarrow y = \frac{dx}{dt} - x$

3) b)

$y = \frac{dx}{dt} - x$

$\Rightarrow \frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt}\right) - \frac{d}{dt}(x)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - \frac{dx}{dt}$

3) c)

$\frac{dy}{dt} = 2y$

$\Rightarrow \frac{d^2x}{dt^2} - \frac{dx}{dt} = 2\left(\frac{dx}{dt} - x\right)$

$\Rightarrow \frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 0$

3) d)

Auxiliary Equation:

$m^2 - 3m + 2 = 0$

$\Rightarrow (m-1)(m-2) = 0$

$\Rightarrow m = 1$ or $m = 2$

General Solution:

$x = Ae^t + Be^{2t}$

3) e)

$x = Ae^t + Be^{2t}$

$\Rightarrow \frac{dx}{dt} = Ae^t + 2Be^{2t}$

Then:

$y = \frac{dx}{dt} - x$

$\Rightarrow y = Ae^t + 2Be^{2t} - (Ae^t + Be^{2t})$

$\Rightarrow y = Be^{2t}$

4) a)

Rearrange the first equation:

$\frac{dx}{dt} = 2x + y$

$\Rightarrow y = \frac{dx}{dt} - 2x$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt}\right) - \frac{d}{dt}(2x)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - 2\frac{dx}{dt}$

Hence:

$\frac{dy}{dt} = x + 2y$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} = x + 2\left(\frac{dx}{dt} - 2x\right)$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} = x + 2\frac{dx}{dt} - 4x$

$\Rightarrow \frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 3x = 0$

4) b)

Auxiliary Equation:

$m^2 - 4m + 3 = 0$

$\Rightarrow (m-1)(m-3) = 0$

$\Rightarrow m = 1$ or $m = 3$

General Solution:

$x = Ae^t + Be^{3t}$

4) c)

$x = Ae^t + Be^{3t}$

$\Rightarrow \frac{dx}{dt} = Ae^t + 3Be^{3t}$

Then:

$y = \frac{dx}{dt} - 2x$

$\Rightarrow y = Ae^t + 3Be^{3t} - 2(Ae^t + Be^{3t})$

$\Rightarrow y = Ae^t + 3Be^{3t} - 2Ae^t - 2Be^{3t}$

$\Rightarrow y = -Ae^t + Be^{3t}$

5) a)

Rearrange the second equation:

$\frac{dy}{dt} = -4x + 5y$

$\Rightarrow x = \frac{5}{4}y - \frac{1}{4}\frac{dy}{dt}$

$\frac{d}{dt}(x) = \frac{d}{dt}\left(\frac{5}{4}y\right) - \frac{d}{dt}\left(\frac{1}{4}\frac{dy}{dt}\right)$

$\Rightarrow \frac{dx}{dt} = \frac{5}{4}\frac{dy}{dt} - \frac{1}{4}\frac{d^2y}{dt^2}$

Hence:

$\frac{dx}{dt} = x + y$

$\Rightarrow \frac{5}{4}\frac{dy}{dt} - \frac{1}{4}\frac{d^2y}{dt^2} = \left(\frac{5}{4}y - \frac{1}{4}\frac{dy}{dt}\right) + y$

$\Rightarrow \frac{5}{4}\frac{dy}{dt} - \frac{1}{4}\frac{d^2y}{dt^2} = \frac{5}{4}y - \frac{1}{4}\frac{dy}{dt} + y$

$\Rightarrow \frac{5}{4}\frac{dy}{dt} - \frac{1}{4}\frac{d^2y}{dt^2} = \frac{9}{4}y - \frac{1}{4}\frac{dy}{dt}$

$\Rightarrow \frac{1}{4}\frac{d^2y}{dt^2} - \frac{3}{2}\frac{dy}{dt} + \frac{9}{4}y = 0$

$\Rightarrow \frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 9y = 0$

5) b)

Auxiliary Equation:

$m^2 - 6m + 9 = 0$

$\Rightarrow (m-3)^2 = 0$

$\Rightarrow m = 3$

General Solution:

$y = Ate^{3t} + Be^{3t}$

5) c)

$y = Ate^{3t} + Be^{3t}$

$\Rightarrow \frac{dy}{dt} = 3Ate^{3t} + Ae^{3t} + 3Be^{3t}$

Then:

$x = \frac{5}{4}y - \frac{1}{4}\frac{dy}{dt}$

$\Rightarrow x = \frac{5}{4}(Ate^{3t} + Be^{3t}) - \frac{1}{4}(3Ate^{3t} + Ae^{3t} + 3Be^{3t})$

$\Rightarrow x = \frac{5}{4}Ate^{3t} + \frac{5}{4}Be^{3t} - \frac{3}{4}Ate^{3t} - \frac{1}{4}Ae^{3t} - \frac{3}{4}Be^{3t}$

$\Rightarrow x = \frac{1}{2}Ate^{3t} - \frac{1}{4}Ae^{3t} + \frac{1}{2}Be^{3t}$

6) a)

$\frac{dx}{dt} = y$

$\Rightarrow y = \frac{dx}{dt}$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt}\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2}$

Hence:

$\frac{dy}{dt} = -x + 2y$

$\Rightarrow \frac{d^2x}{dt^2} = -x + 2\frac{dx}{dt}$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = 0$

6) b)

Auxiliary Equation:

$m^2 - 2m + 1 = 0$

$\Rightarrow (m-1)^2 = 0$

$\Rightarrow m = 1$

General Solution:

$x = Ate^t + Be^t$

$x = Ate^t + Be^t$

$\Rightarrow \frac{dx}{dt} = Ate^t + Ae^t + Be^t$

Then:

$y = \frac{dx}{dt}$

$\Rightarrow y = Ate^t + Ae^t + Be^t$

6) c)

$x = Ate^t + Be^t$

Substitute $t = 0$ and $x = 1$:

$1 = A \times 0 \times e^0 + Be^0$

$\Rightarrow B = 1$

$y = Ate^t + Ae^t + e^t$

Substitute $t = 0$ and $y = 0$:

$0 = A \times 0 \times e^0 + Ae^0 + e^0$

$\Rightarrow 0 = A + 1$

$\Rightarrow A = -1$

$\therefore x = -te^t + e^t$ and $y = -te^t$

7) a)

Rearrange the second equation:

$\frac{dy}{dt} = x - y$

$\Rightarrow x = \frac{dy}{dt} + y$

$\frac{d}{dt}(x) = \frac{d}{dt}\left(\frac{dy}{dt}\right) + \frac{d}{dt}(y)$

$\Rightarrow \frac{dx}{dt} = \frac{d^2y}{dt^2} + \frac{dy}{dt}$

Hence:

$\frac{dx}{dt} = x + 3y$

$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} = \left(\frac{dy}{dt} + y\right) + 3y$

$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} = \frac{dy}{dt} + y + 3y$

$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} = \frac{dy}{dt} + 4y$

$\Rightarrow \frac{d^2y}{dt^2} - 4y = 0$

7) b)

Auxiliary Equation:

$m^2 - 4 = 0$

$\Rightarrow (m-2)(m+2) = 0$

$\Rightarrow m = 2$ or $m = -2$

General Solution:

$y = Ae^{2t} + Be^{-2t}$

7) c)

$y = Ae^{2t} + Be^{-2t}$

$\Rightarrow \frac{dy}{dt} = 2Ae^{2t} - 2Be^{-2t}$

Then:

$x = \frac{dy}{dt} + y$

$\Rightarrow x = (2Ae^{2t} - 2Be^{-2t}) + (Ae^{2t} + Be^{-2t})$

$\Rightarrow x = 2Ae^{2t} - 2Be^{-2t} + Ae^{2t} + Be^{-2t}$

$\Rightarrow x = 3Ae^{2t} - Be^{-2t}$

7) d)

$x = 3Ae^{2t} - Be^{-2t}$

Substitute $t = 0$ and $x = 4$:

$4 = 3Ae^{0} - Be^{0}$

$\Rightarrow 3A - B = 4$ (1)

$y = Ae^{2t} + Be^{-2t}$

Substitute $t = 0$ and $y = 0$:

$0 = Ae^{0} + Be^{0}$

$\Rightarrow A + B = 0$ (2)

Solve (1) & (2)

$A = 1$ and $B = -1$

$\therefore x = 3e^{2t} + e^{-2t}$ and $y = e^{2t} - e^{-2t}$

8) a)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + y$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 2x\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = x + 2y$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 2x\right)$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2\frac{\mathrm{d}x}{\mathrm{d}t} - 4x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 0$

Auxiliary Equation:

$m^2 - 4m + 3 = 0$

$\Rightarrow (m-1)(m-3) = 0$

$\Rightarrow m = 1$ or $m = 3$

General Solution:

$x = A\mathrm{e}^t + B\mathrm{e}^{3t}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = A\mathrm{e}^t + 3B\mathrm{e}^{3t}$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x$

$\Rightarrow y = A\mathrm{e}^t + 3B\mathrm{e}^{3t} - 2(A\mathrm{e}^t + B\mathrm{e}^{3t})$

$\Rightarrow y = A\mathrm{e}^t + 3B\mathrm{e}^{3t} - 2A\mathrm{e}^t - 2B\mathrm{e}^{3t}$

$\Rightarrow y = -A\mathrm{e}^t + B\mathrm{e}^{3t}$

So:

$x = A\mathrm{e}^t + B\mathrm{e}^{3t}$ and $y = -A\mathrm{e}^t + B\mathrm{e}^{3t}$

8) b)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + 2y$

$\Rightarrow y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - x$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - x\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -2x + 6y$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} = -2x + 6\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - x\right)$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} = -2x + 3\frac{\mathrm{d}x}{\mathrm{d}t} - 6x$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 8x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 8\frac{\mathrm{d}x}{\mathrm{d}t} + 16x = 0$

Auxiliary Equation:

$m^2 - 8m + 16 = 0$

$\Rightarrow (m-4)^2 = 0$

$\Rightarrow m = 4$

General Solution:

$x = At\mathrm{e}^{4t} + B\mathrm{e}^{4t}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = 4At\mathrm{e}^{4t} + A\mathrm{e}^{4t} + 4B\mathrm{e}^{4t}$

Then:

$y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - x$

$\Rightarrow y = \frac{1}{2}(4At\mathrm{e}^{4t} + A\mathrm{e}^{4t} + 4B\mathrm{e}^{4t}) - (At\mathrm{e}^{4t} + B\mathrm{e}^{4t})$

$\Rightarrow y = 2At\mathrm{e}^{4t} + \frac{1}{2}A\mathrm{e}^{4t} + 2B\mathrm{e}^{4t} - At\mathrm{e}^{4t} - B\mathrm{e}^{4t}$

$\Rightarrow y = At\mathrm{e}^{4t} + \frac{1}{2}A\mathrm{e}^{4t} + B\mathrm{e}^{4t}$

So:

$x = At\mathrm{e}^{4t} + B\mathrm{e}^{4t}$ and $y = At\mathrm{e}^{4t} + \frac{1}{2}A\mathrm{e}^{4t} + B\mathrm{e}^{4t}$

8) c)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2y$

$\Rightarrow y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}x$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}x\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -x - y$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} = -x - \left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}x\right)$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} = -x - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{2}x$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{2}x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x = 0$

Auxiliary Equation:

$m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm\mathrm{i}$

General Solution:

$x = A\cos t + B\sin t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -A\sin t + B\cos t$

Then:

$y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}x$

$\Rightarrow y = \frac{1}{2}(-A\sin t + B\cos t) - \frac{1}{2}(A\cos t + B\sin t)$

$\Rightarrow y = -\frac{1}{2}A\sin t + \frac{1}{2}B\cos t - \frac{1}{2}A\cos t - \frac{1}{2}B\sin t$

$\Rightarrow y = -\frac{1}{2}(A + B)\sin t + \frac{1}{2}(-A + B)\cos t$

So:

$x = A\cos t + B\sin t$ and $y = -\frac{1}{2}(A + B)\sin t + \frac{1}{2}(-A + B)\cos t$

8) d)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + y$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 2x\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -5x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} = -5x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$

Auxiliary Equation:

$m^2 - 2m + 5 = 0$

$\Rightarrow (m-1)^2 - 1 + 5 = 0$

$\Rightarrow (m-1)^2 = -4$

$\Rightarrow m = 1 \pm 2\mathrm{i}$

General Solution:

$x = \mathrm{e}^t(A\cos 2t + B\sin 2t)$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^t(-2A\sin 2t + 2B\cos 2t) + \mathrm{e}^t(A\cos 2t + B\sin 2t)$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x$

$\Rightarrow y = \mathrm{e}^t(-2A\sin 2t + 2B\cos 2t) + \mathrm{e}^t(A\cos 2t + B\sin 2t) - 2\mathrm{e}^t(A\cos 2t + B\sin 2t)$

$\Rightarrow y = -2A\mathrm{e}^t\sin 2t + 2B\mathrm{e}^t\cos 2t + A\mathrm{e}^t\cos 2t + B\mathrm{e}^t\sin 2t - 2A\mathrm{e}^t\cos 2t - 2B\mathrm{e}^t\sin 2t$

$\Rightarrow y = -2A\mathrm{e}^t\sin 2t + 2B\mathrm{e}^t\cos 2t - A\mathrm{e}^t\cos 2t - B\mathrm{e}^t\sin 2t$

$\Rightarrow y = (-2A - B)\mathrm{e}^t\sin 2t + (-A + 2B)\mathrm{e}^t\cos 2t$

So:

$x = \mathrm{e}^t(A\cos 2t + B\sin 2t)$ and $y = (-2A - B)\mathrm{e}^t\sin 2t + (-A + 2B)\mathrm{e}^t\cos 2t$

9) a)

Rearrange the first equation:

$\frac{dx}{dt} = x + y + 1$

$\Rightarrow y = \frac{dx}{dt} - x - 1$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt} - x - 1\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - \frac{dx}{dt}$

Hence:

$\frac{dy}{dt} = x + y - 1$

$\Rightarrow \frac{d^2x}{dt^2} - \frac{dx}{dt} = x + \left(\frac{dx}{dt} - x - 1\right) - 1$

$\Rightarrow \frac{d^2x}{dt^2} - \frac{dx}{dt} = x + \frac{dx}{dt} - x - 1 - 1$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} = -2$

9) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

Auxiliary Equation:

$m^2 - 2m = 0$

$\Rightarrow (m-2)m = 0$

$\Rightarrow m = 2$ or $m = 0$

Complementary Function:

$x = A\mathrm{e}^{2t} + B$

Try this Particular Integral:

$x = pt$

$\frac{\mathrm{d}x}{\mathrm{d}t} = p$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} = -2$

$\Rightarrow (0) - 2(p) = -2$

$\Rightarrow -2p = -2$

$\Rightarrow p = 1$

Particular Integral:

$x = t$

General Solution:

$x = A\mathrm{e}^{2t} + B + t$

9) c)

$x = A\mathrm{e}^{2t} + B + t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = 2A\mathrm{e}^{2t} + 1$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - x - 1$

$\Rightarrow y = (2A\mathrm{e}^{2t} + 1) - (A\mathrm{e}^{2t} + B + t) - 1$

$\Rightarrow y = 2A\mathrm{e}^{2t} + 1 - A\mathrm{e}^{2t} - B - t - 1$

$\Rightarrow y = A\mathrm{e}^{2t} - B - t$

9) d)

$x = Ae^{2t} + B + t$

Substitute $t = 0$ and $x = 1$:

$1 = Ae^{0} + B + 0$

$\Rightarrow A + B = 1$ (1)

$y = Ae^{2t} - B - t$

Substitute $t = 0$ and $y = 0$:

$0 = Ae^{0} - B - 0$

$\Rightarrow A - B = 0$ (2)

Solve (1) & (2)

$A = \frac{1}{2}$ and $B = \frac{1}{2}$

$\therefore x = \frac{1}{2}e^{2t} + \frac{1}{2} + t$ and $y = \frac{1}{2}e^{2t} - \frac{1}{2} - t$

10) a)

Rearrange the first equation:

$\frac{dx}{dt} = 2y + 6$

$\Rightarrow y = \frac{1}{2}\frac{dx}{dt} - 3$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{1}{2}\frac{dx}{dt} - 3\right)$

$\Rightarrow \frac{dy}{dt} = \frac{1}{2}\frac{d^2x}{dt^2}$

Hence:

$\frac{dy}{dt} = 2x - 4$

$\Rightarrow \frac{1}{2}\frac{d^2x}{dt^2} = 2x - 4$

$\Rightarrow \frac{1}{2}\frac{d^2x}{dt^2} - 2x = -4$

$\Rightarrow \frac{d^2x}{dt^2} - 4x = -8$

Homogeneous Equation:

$\frac{d^2x}{dt^2} - 4x = 0$

Auxiliary Equation:

$m^2 - 4 = 0$

$\Rightarrow (m - 2)(m + 2) = 0$

$\Rightarrow m = 2$ or $m = -2$

Complementary Function:

$x = Ae^{2t} + Be^{-2t}$

Try this Particular Integral:

$x = k$

$\frac{dx}{dt} = 0$

$\frac{d^2x}{dt^2} = 0$

$\frac{d^2x}{dt^2} - 4x = -8$

$\Rightarrow (0) - 4(k) = -8$

$\Rightarrow -4k = -8$

$\Rightarrow k = 2$

Particular Integral:

$x = 2$

General Solution:

$x = Ae^{2t} + Be^{-2t} + 2$

$\Rightarrow \frac{dx}{dt} = 2Ae^{2t} - 2Be^{-2t}$

Then:

$y = \frac{1}{2}\frac{dx}{dt} - 3$

$\Rightarrow y = \frac{1}{2}(2Ae^{2t} - 2Be^{-2t}) - 3$

$\Rightarrow y = Ae^{2t} - Be^{-2t} - 3$

So:

$x = Ae^{2t} + Be^{-2t} + 2$ and $y = Ae^{2t} - Be^{-2t} - 3$

10) b)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3x + y + 2$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 2$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 2\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -x + y - 2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} = -x + \left(\frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 2\right) - 2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} = -x + \frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 2 - 2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = -4$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 0$

Auxiliary Equation:

$m^2 - 4m + 4 = 0$

$\Rightarrow (m-2)^2 = 0$

$\Rightarrow m = 2$

Complementary Function:

$x = At\mathrm{e}^{2t} + B\mathrm{e}^{2t}$

Try this Particular Integral:

$x = k$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = -4$

$\Rightarrow (0) - 4(0) + 4(k) = -4$

$\Rightarrow 4k = -4$

$\Rightarrow k = -1$

Particular Integral:

$x = -1$

General Solution:

$x = Ate^{2t} + Be^{2t} - 1$

$\Rightarrow \frac{dx}{dt} = 2Ate^{2t} + Ae^{2t} + 2Be^{2t}$

Then:

$y = \frac{dx}{dt} - 3x - 2$

$\Rightarrow y = (2Ate^{2t} + Ae^{2t} + 2Be^{2t}) - 3(Ate^{2t} + Be^{2t} - 1) - 2$

$\Rightarrow y = 2Ate^{2t} + Ae^{2t} + 2Be^{2t} - 3Ate^{2t} - 3Be^{2t} + 3 - 2$

$\Rightarrow y = -Ate^{2t} + Ae^{2t} - Be^{2t} + 1$

$\Rightarrow y = -Ate^{2t} + (A - B)e^{2t} + 1$

So:

$x = Ate^{2t} + Be^{2t} - 1$ and $y = -Ate^{2t} + (A - B)e^{2t} + 1$

10) c)

Rearrange the first equation:

$\frac{dx}{dt} = 2x + y + 3$

$\Rightarrow y = \frac{dx}{dt} - 2x - 3$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt} - 2x - 3\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - 2\frac{dx}{dt}$

Hence:

$\frac{dy}{dt} = -13x - 2y - 5$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} = -13x - 2\left(\frac{dx}{dt} - 2x - 3\right) - 5$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} = -13x - 2\frac{dx}{dt} + 4x + 6 - 5$

$\Rightarrow \frac{d^2x}{dt^2} + 9x = 1$

Homogeneous Equation:

$\frac{d^2x}{dt^2} + 9x = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3i$

Complementary Function:

$x = A\cos 3t + B\sin 3t$

Try this Particular Integral:

$x = k$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = 1$

$\Rightarrow (0) + 9(k) = 1$

$\Rightarrow 9k = 1$

$\Rightarrow k = \frac{1}{9}$

Particular Integral:

$x = \frac{1}{9}$

General Solution:

$x = A\cos 3t + B\sin 3t + \frac{1}{9}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -3A\sin 3t + 3B\cos 3t$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x - 3$

$\Rightarrow y = (-3A\sin 3t + 3B\cos 3t) - 2\left(A\cos 3t + B\sin 3t + \frac{1}{9}\right) - 3$

$\Rightarrow y = -3A\sin 3t + 3B\cos 3t - 2A\cos 3t - 2B\sin 3t - \frac{2}{9} - 3$

$\Rightarrow y = (-3A - 2B)\sin 3t + (3B - 2A)\cos 3t - \frac{29}{9}$

So:

$x = A\cos 3t + B\sin 3t + \frac{1}{9}$ and $y = (-3A - 2B)\sin 3t + (3B - 2A)\cos 3t - \frac{29}{9}$

10) d)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3x + 2y + 1$

$\Rightarrow y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -2x + 3y + 1$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} = -2x + 3\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\right) + 1$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} = -2x + \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{9}{2}x - \frac{3}{2} + 1$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{13}{2}x = -\frac{1}{2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = -1$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = 0$

Auxiliary Equation:

$m^2 - 6m + 13 = 0$

$\Rightarrow (m-3)^2 - 9 + 13 = 0$

$\Rightarrow (m-3)^2 = -4$

$\Rightarrow m = 3 \pm 2\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{3t}(A\cos 2t + B\sin 2t)$

Try this Particular Integral:

$x = k$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = -1$

$\Rightarrow (0) - 6(0) + 13(k) = -1$

$\Rightarrow 13k = -1$

$\Rightarrow k = -\frac{1}{13}$

Particular Integral:

$x = -\frac{1}{13}$

General Solution:

$x = \mathrm{e}^{3t}(A\cos 2t + B\sin 2t) - \frac{1}{13}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{3t}(-2A\sin 2t + 2B\cos 2t) + 3\mathrm{e}^{3t}(A\cos 2t + B\sin 2t)$

Then:

$y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}$

$\Rightarrow y = \frac{1}{2}\left(\mathrm{e}^{3t}(-2A\sin 2t + 2B\cos 2t) + 3\mathrm{e}^{3t}(A\cos 2t + B\sin 2t)\right) - \frac{3}{2}\left(\mathrm{e}^{3t}(A\cos 2t + B\sin 2t) - \frac{1}{13}\right) - \frac{1}{2}$

$\Rightarrow y = \frac{1}{2}\mathrm{e}^{3t}(-2A\sin 2t + 2B\cos 2t) + \frac{3}{2}\mathrm{e}^{3t}(A\cos 2t + B\sin 2t) - \frac{3}{2}\mathrm{e}^{3t}(A\cos 2t + B\sin 2t) + \frac{3}{26} - \frac{1}{2}$

$\Rightarrow y = \frac{1}{2}\mathrm{e}^{3t}(-2A\sin 2t + 2B\cos 2t) - \frac{5}{13}$

$\Rightarrow y = \mathrm{e}^{3t}(B\cos 2t - A\sin 2t) - \frac{5}{13}$

So:

$x = \mathrm{e}^{3t}(A\cos 2t + B\sin 2t) - \frac{1}{13}$ and $y = \mathrm{e}^{3t}(B\cos 2t - A\sin 2t) - \frac{5}{13}$

11)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = x + y + t$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - x - t$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - x - t\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 1$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = 2y + t^2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 1 = 2\left(\frac{\mathrm{d}x}{\mathrm{d}t} - x - t\right) + t^2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 1 = 2\frac{\mathrm{d}x}{\mathrm{d}t} - 2x - 2t + t^2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 1 - 2t + t^2$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0$

Auxiliary Equation:

$m^2 - 3m + 2 = 0$

$\Rightarrow (m-1)(m-2) = 0$

$\Rightarrow m = 1$ or $m = 2$

Complementary Function:

$x = A\mathrm{e}^t + B\mathrm{e}^{2t}$

Try this Particular Integral:

$x = pt^2 + qt + r$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2pt + q$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 2p$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 1 - 2t + t^2$

$\Rightarrow (2p) - 3(2pt + q) + 2(pt^2 + qt + r) = 1 - 2t + t^2$

$\Rightarrow 2p - 6pt - 3q + 2pt^2 + 2qt + 2r = 1 - 2t + t^2$

$\Rightarrow (2p - 3q + 2r) + (-6p + 2q)t + (2p)t^2 = 1 - 2t + t^2$

Compare coefficients

$2p = 1 \Rightarrow p = \frac{1}{2}$

$-6 \times \frac{1}{2} + 2q = -2 \Rightarrow q = \frac{1}{2}$

$2 \times \frac{1}{2} - 3 \times \frac{1}{2} + 2r = 1 \Rightarrow r = \frac{3}{4}$

Particular Integral:

$x = \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}$

General Solution:

$x = Ae^t + Be^{2t} + \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}$

$\Rightarrow \frac{dx}{dt} = Ae^t + 2Be^{2t} + t + \frac{1}{2}$

Then:

$y = \frac{dx}{dt} - x - t$

$\Rightarrow y = \left(Ae^t + 2Be^{2t} + t + \frac{1}{2}\right) - \left(Ae^t + Be^{2t} + \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}\right) - t$

$\Rightarrow y = Ae^t + 2Be^{2t} + t + \frac{1}{2} - Ae^t - Be^{2t} - \frac{1}{2}t^2 - \frac{1}{2}t - \frac{3}{4} - t$

$\Rightarrow y = Be^{2t} - \frac{1}{2}t^2 - \frac{1}{2}t - \frac{1}{4}$

So:

$x = Ae^t + Be^{2t} + \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}$ and $y = Be^{2t} - \frac{1}{2}t^2 - \frac{1}{2}t - \frac{1}{4}$

$x = Ae^t + Be^{2t} + \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}$

Substitute $t = 0$ and $x = \frac{7}{4}$:

$\frac{7}{4} = Ae^0 + Be^0 + \frac{1}{2} \times 0^2 + \frac{1}{2} \times 0 + \frac{3}{4}$

$\Rightarrow \frac{7}{4} = A + B + \frac{3}{4}$

$\Rightarrow A + B = 1$ (1)

$y = Be^{2t} - \frac{1}{2}t^2 - \frac{1}{2}t - \frac{1}{4}$

Substitute $t = 0$ and $y = \frac{1}{4}$:

$\frac{1}{4} = Be^0 - \frac{1}{2} \times 0^2 - \frac{1}{2} \times 0 - \frac{1}{4}$

$\Rightarrow \frac{1}{4} = B - \frac{1}{4}$

$\Rightarrow B = \frac{1}{2}$

$\therefore A + \frac{1}{2} = 1 \Rightarrow A = \frac{1}{2}$

$\therefore x = \frac{1}{2}e^t + \frac{1}{2}e^{2t} + \frac{1}{2}t^2 + \frac{1}{2}t + \frac{3}{4}$ and $y = \frac{1}{2}e^{2t} - \frac{1}{2}t^2 - \frac{1}{2}t - \frac{1}{4}$

12) a)

Rearrange the first equation:

$\frac{dx}{dt} = 2x + y + \sin t$

$\Rightarrow y = \frac{dx}{dt} - 2x - \sin t$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt} - 2x - \sin t\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - \cos t$

Hence:

$\frac{dy}{dt} = -x + \cos t$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - \cos t = -x + \cos t$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = 2\cos t$

12) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} + x = 0$

Auxiliary Equation:

$m^2 - 2m + 1 = 0$

$\Rightarrow (m-1)^2 = 0$

$\Rightarrow m = 1$

Complementary Function:

$x = At\mathrm{e}^t + B\mathrm{e}^t$

Try this Particular Integral:

$x = p\cos t + q\sin t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -p\sin t + q\cos t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -p\cos t - q\sin t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} + x = 2\cos t$

$\Rightarrow (-p\cos t - q\sin t) - 2(-p\sin t + q\cos t) + (p\cos t + q\sin t) = 2\cos t$

$\Rightarrow -p\cos t - q\sin t + 2p\sin t - 2q\cos t + p\cos t + q\sin t = 2\cos t$

$\Rightarrow 2p\sin t - 2q\cos t = 2\cos t$

Compare coefficients

$2p = 0 \Rightarrow p = 0$

$-2q = 2 \Rightarrow q = -1$

Particular Integral:

$x = -\sin t$

General Solution:

$x = At\mathrm{e}^t + B\mathrm{e}^t - \sin t$

12) c)

$x = At\mathrm{e}^t + B\mathrm{e}^t - \sin t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = At\mathrm{e}^t + A\mathrm{e}^t + B\mathrm{e}^t - \cos t$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin t$

$\Rightarrow y = (At\mathrm{e}^t + A\mathrm{e}^t + B\mathrm{e}^t - \cos t) - 2(At\mathrm{e}^t + B\mathrm{e}^t - \sin t) - \sin t$

$\Rightarrow y = At\mathrm{e}^t + A\mathrm{e}^t + B\mathrm{e}^t - \cos t - 2At\mathrm{e}^t - 2B\mathrm{e}^t + 2\sin t - \sin t$

$\Rightarrow y = -At\mathrm{e}^t + (A-B)\mathrm{e}^t - \cos t + \sin t$

12) d)

$x = Ate^t + Be^t - \sin t$

Substitute $t = 0$ and $x = 1$:

$1 = A \times 0 \times e^0 + Be^0 - \sin 0$

$\Rightarrow B = 1$

$y = -Ate^t + (A - B)e^t - \cos t + \sin t$

Substitute $t = 0$ and $y = 1$:

$1 = -A \times 0 \times e^0 + (A - 1)e^0 - \cos 0 + \sin 0$

$\Rightarrow 1 = A - 1 - 1$

$\Rightarrow A = 3$

$\therefore x = 3te^t + e^t - \sin t$ and $y = -3te^t + 2e^t - \cos t + \sin t$

13) a)

Rearrange the first equation:

$\frac{dx}{dt} = 2x + y + \sin t$

$\Rightarrow y = \frac{dx}{dt} - 2x - \sin t$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{dx}{dt} - 2x - \sin t\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - \cos t$

Hence:

$\frac{dy}{dt} = 2x + 3y + \cos t$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - \cos t = 2x + 3\left(\frac{dx}{dt} - 2x - \sin t\right) + \cos t$

$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} - \cos t = 2x + 3\frac{dx}{dt} - 6x - 3\sin t + \cos t$

$\Rightarrow \frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 4x = -3\sin t + 2\cos t$

Homogeneous Equation:

$\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 4x = 0$

Auxiliary Equation:

$m^2 - 5m + 4 = 0$

$\Rightarrow (m - 1)(m - 4) = 0$

$\Rightarrow m = 1$ or $m = 4$

Complementary Function:

$x = Ae^t + Be^{4t}$

Try this Particular Integral:

$x = p\cos t + q\sin t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -p\sin t + q\cos t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -p\cos t - q\sin t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 5\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = -3\sin t + 2\cos t$

$\Rightarrow (-p\cos t - q\sin t) - 5(-p\sin t + q\cos t) + 4(p\cos t + q\sin t) = -3\sin t + 2\cos t$

$\Rightarrow -p\cos t - q\sin t + 5p\sin t - 5q\cos t + 4p\cos t + 4q\sin t = -3\sin t + 2\cos t$

$\Rightarrow (-p - 5q + 4p)\cos t + (-q + 5p + 4q)\sin t = -3\sin t + 2\cos t$

$\Rightarrow (3p - 5q)\cos t + (5p + 3q)\sin t = -3\sin t + 2\cos t$

Compare coefficients

$3p - 5q = 2$ (1)

$5p + 3q = -3$ (2)

Solve (1) & (2)

$p = -\frac{9}{34}$ and $q = -\frac{19}{34}$

Particular Integral:

$x = -\frac{9}{34}\cos t - \frac{19}{34}\sin t$

General Solution:

$x = A\mathrm{e}^t + B\mathrm{e}^{4t} - \frac{9}{34}\cos t - \frac{19}{34}\sin t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = A\mathrm{e}^t + 4B\mathrm{e}^{4t} + \frac{9}{34}\sin t - \frac{19}{34}\cos t$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin t$

$\Rightarrow y = \left(A\mathrm{e}^t + 4B\mathrm{e}^{4t} + \frac{9}{34}\sin t - \frac{19}{34}\cos t\right) - 2\left(A\mathrm{e}^t + B\mathrm{e}^{4t} - \frac{9}{34}\cos t - \frac{19}{34}\sin t\right) - \sin t$

$\Rightarrow y = A\mathrm{e}^t + 4B\mathrm{e}^{4t} + \frac{9}{34}\sin t - \frac{19}{34}\cos t - 2A\mathrm{e}^t - 2B\mathrm{e}^{4t} + \frac{9}{17}\cos t + \frac{19}{17}\sin t - \sin t$

$\Rightarrow y = -A\mathrm{e}^t + 2B\mathrm{e}^{4t} + \frac{13}{34}\sin t - \frac{1}{34}\cos t$

So:

$x = A\mathrm{e}^t + B\mathrm{e}^{4t} - \frac{9}{34}\cos t - \frac{19}{34}\sin t$ and $y = -A\mathrm{e}^t + 2B\mathrm{e}^{4t} + \frac{13}{34}\sin t - \frac{1}{34}\cos t$

13) b)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = x + y + \sin 4t$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - x - \sin 4t$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - x - \sin 4t\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 4\cos 4t$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -x + 3y + 3\sin 4t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 4\cos 4t = -x + 3\left(\frac{\mathrm{d}x}{\mathrm{d}t} - x - \sin 4t\right) + 3\sin 4t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{\mathrm{d}x}{\mathrm{d}t} - 4\cos 4t = -x + 3\frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 3\sin 4t + 3\sin 4t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 4\cos 4t$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 0$

Auxiliary Equation:

$m^2 - 4m + 4 = 0$

$\Rightarrow (m-2)^2 = 0$

$\Rightarrow m = 2$

Complementary Function:

$x = Ate^{2t} + Be^{2t}$

Try this Particular Integral:

$x = p\cos 4t + q\sin 4t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -4p\sin 4t + 4q\cos 4t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -16p\cos 4t - 16q\sin 4t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 4\cos 4t$

$\Rightarrow (-16p\cos 4t - 16q\sin 4t) - 4(-4p\sin 4t + 4q\cos 4t) + 4(p\cos 4t + q\sin 4t) = 4\cos 4t$

$\Rightarrow -16p\cos 4t - 16q\sin 4t + 16p\sin 4t - 16q\cos 4t + 4p\cos 4t + 4q\sin 4t = 4\cos 4t$

$\Rightarrow (-16p - 16q + 4p)\cos 4t + (-16q + 16p + 4q)\sin 4t = 4\cos 4t$

$\Rightarrow (-12p - 16q)\cos 4t + (16p - 12q)\sin 4t = 4\cos 4t$

Compare coefficients

$-12p - 16q = 4$ (1)

$16p - 12q = 0$ (2)

Solve (1) & (2)

$p = -\frac{3}{25}$ and $q = -\frac{4}{25}$

Particular Integral:

$x = -\frac{3}{25}\cos 4t - \frac{4}{25}\sin 4t$

General Solution:

$x = Ate^{2t} + Be^{2t} - \frac{3}{25}\cos 4t - \frac{4}{25}\sin 4t$

$\Rightarrow \frac{dx}{dt} = 2Ate^{2t} + Ae^{2t} + 2Be^{2t} + \frac{12}{25}\sin 4t - \frac{16}{25}\cos 4t$

Then:

$y = \frac{dx}{dt} - x - \sin 4t$

$\Rightarrow y = \left(2Ate^{2t} + Ae^{2t} + 2Be^{2t} + \frac{12}{25}\sin 4t - \frac{16}{25}\cos 4t\right) - \left(Ate^{2t} + Be^{2t} - \frac{3}{25}\cos 4t - \frac{4}{25}\sin 4t\right) - \sin 4t$

$\Rightarrow y = 2Ate^{2t} + Ae^{2t} + 2Be^{2t} + \frac{12}{25}\sin 4t - \frac{16}{25}\cos 4t - Ate^{2t} - Be^{2t} + \frac{3}{25}\cos 4t + \frac{4}{25}\sin 4t - \sin 4t$

$\Rightarrow y = Ate^{2t} + (A + B)e^{2t} - \frac{13}{25}\cos 4t - \frac{9}{25}\sin 4t$

So:

$x = Ate^{2t} + Be^{2t} - \frac{3}{25}\cos 4t - \frac{4}{25}\sin 4t$

and $y = Ate^{2t} + (A + B)e^{2t} - \frac{13}{25}\cos 4t - \frac{9}{25}\sin 4t$

13) c)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3x + 2y + \cos 2t$

$\Rightarrow y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\cos 2t$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\cos 2t\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} + \sin 2t$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -9x - 3y - 5\cos 2t$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} + \sin 2t = -9x - 3\left(\frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\cos 2t\right) - 5\cos 2t$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} + \sin 2t = -9x - \frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{9}{2}x + \frac{3}{2}\cos 2t - 5\cos 2t$

$\Rightarrow \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{9}{2}x = -\frac{7}{2}\cos 2t - \sin 2t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = -7\cos 2t - 2\sin 2t$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3\mathrm{i}$

Complementary Function:

$x = A\cos 3t + B\sin 3t$

Try this Particular Integral:

$x = p\cos 2t + q\sin 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2p\sin 2t + 2q\cos 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -4p\cos 2t - 4q\sin 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = -7\cos 2t - \frac{1}{2}\sin 2t$

$\Rightarrow (-4p\cos 2t - 4q\sin 2t) + 9(p\cos 2t + q\sin 2t) = -7\cos 2t - 2\sin 2t$

$\Rightarrow -4p\cos 2t - 4q\sin 2t + 9p\cos 2t + 9q\sin 2t = -7\cos 2t - 2\sin 2t$

$\Rightarrow 5p\cos 2t + 5q\sin 2t = -7\cos 2t - 2\sin 2t$

Compare coefficients

$5p = -7 \Rightarrow p = -\frac{7}{5}$

$5q = -2 \Rightarrow q = -\frac{2}{5}$

Particular Integral:

$x = -\frac{7}{5}\cos 2t - \frac{2}{5}\sin 2t$

General Solution:

$x = A\cos 3t + B\sin 3t - \frac{7}{5}\cos 2t - \frac{2}{5}\sin 2t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -3A\sin 3t + 3B\cos 3t + \frac{14}{5}\sin 2t - \frac{4}{5}\cos 2t$

Then:

$y = \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2}x - \frac{1}{2}\cos 2t$

$\Rightarrow y = \frac{1}{2}\left(-3A\sin 3t + 3B\cos 3t + \frac{14}{5}\sin 2t - \frac{4}{5}\cos 2t\right) - \frac{3}{2}\left(A\cos 3t + B\sin 3t - \frac{7}{5}\cos 2t - \frac{2}{5}\sin 2t\right) - \frac{1}{2}\cos 2t$

$\Rightarrow y = -\frac{3}{2}A\sin 3t + \frac{3}{2}B\cos 3t + \frac{7}{5}\sin 2t - \frac{2}{5}\cos 2t - \frac{3}{2}A\cos 3t - \frac{3}{2}B\sin 3t + \frac{21}{10}\cos 2t + \frac{3}{5}\sin 2t - \frac{1}{2}\cos 2t$

$\Rightarrow y = \frac{3}{2}(-A + B)\cos 3t - \frac{3}{2}(A + B)\sin 3t + \frac{6}{5}\cos 2t + 2\sin 2t$

So:

$x = A\cos 3t + B\sin 3t - \frac{7}{5}\cos 2t - \frac{2}{5}\sin 2t$

and $y = \frac{3}{2}(-A + B)\cos 3t - \frac{3}{2}(A + B)\sin 3t + \frac{6}{5}\cos 2t + 2\sin 2t$

13) d)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + y + \sin 5t$

$\Rightarrow y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin 5t$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin 5t\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 5\cos 5t$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = -x + 2y - 25\cos 5t + 22\sin 5t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 5\cos 5t = -x + 2\left(\frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin 5t\right) - 25\cos 5t + 22\sin 5t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 5\cos 5t = -x + 2\frac{\mathrm{d}x}{\mathrm{d}t} - 4x - 2\sin 5t - 25\cos 5t + 22\sin 5t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = -20\cos 5t + 20\sin 5t$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$

Auxiliary Equation:

$m^2 - 4m + 5 = 0$

$\Rightarrow (m-2)^2 - 4 + 5 = 0$

$\Rightarrow (m-2)^2 = -1$

$\Rightarrow m = 2 \pm \mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{2t}(A\cos t + B\sin t)$

Try this Particular Integral:

$x = p\cos 5t + q\sin 5t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -5p\sin 5t + 5q\cos 5t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -25p\cos 5t - 25q\sin 5t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = -20\cos 5t + 20\sin 5t$

$\Rightarrow (-25p\cos 5t - 25q\sin 5t) - 4(-5p\sin 5t + 5q\cos 5t) + 5(p\cos 5t + q\sin 5t) = -20\cos 5t + 20\sin 5t$

$\Rightarrow -25p\cos 5t - 25q\sin 5t + 20p\sin 5t - 20q\cos 5t + 5p\cos 5t + 5q\sin 5t = -20\cos 5t + 20\sin 5t$

$\Rightarrow (-25p - 20q + 5p)\cos 5t + (-25q + 20p + 5q)\sin 5t = -20\cos 5t + 20\sin 5t$

$\Rightarrow (-20p - 20q)\cos 5t + (20p - 20q)\sin 5t = -20\cos 5t + 20\sin 5t$

Compare coefficients

$-20p - 20q = -20$ (1)

$20p - 20q = 20$ (2)

Solve (1) & (2)

$p = 1$ and $q = 0$

Particular Integral:

$x = \cos 5t$

General Solution:

$x = \mathrm{e}^{2t}(A\cos t + B\sin t) + \cos 5t$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{2t}(-A\sin t + B\cos t) + 2\mathrm{e}^{2t}(A\cos t + B\sin t) - 5\sin 5t$

Then:

$y = \frac{\mathrm{d}x}{\mathrm{d}t} - 2x - \sin 5t$

$\Rightarrow y = (\mathrm{e}^{2t}(-A\sin t + B\cos t) + 2\mathrm{e}^{2t}(A\cos t + B\sin t) - 5\sin 5t) - 2(\mathrm{e}^{2t}(A\cos t + B\sin t) + \cos 5t) - \sin 5t$

$\Rightarrow y = \mathrm{e}^{2t}(-A\sin t + B\cos t) + 2\mathrm{e}^{2t}(A\cos t + B\sin t) - 5\sin 5t - 2\mathrm{e}^{2t}(A\cos t + B\sin t) - 2\cos 5t - \sin 5t$

$\Rightarrow y = \mathrm{e}^{2t}(-A\sin t + B\cos t + 2A\cos t + 2B\sin t - 2A\cos t - 2B\sin t) - 2\cos 5t - 6\sin 5t$

$\Rightarrow y = \mathrm{e}^{2t}(B\cos t - A\sin t) - 2\cos 5t - 6\sin 5t$

So:

$x = \mathrm{e}^{2t}(A\cos t + B\sin t) + \cos 5t$

and $y = \mathrm{e}^{2t}(B\cos t - A\sin t) - 2\cos 5t - 6\sin 5t$

14)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + 3y + \mathrm{e}^{-2t}$

$\Rightarrow y = \frac{1}{3}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{2}{3}x - \frac{1}{3}\mathrm{e}^{-2t}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{3}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{2}{3}x - \frac{1}{3}\mathrm{e}^{-2t}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{3}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{3}\mathrm{e}^{-2t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = x + 2y + \mathrm{e}^{-2t}$

$\Rightarrow \frac{1}{3}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{3}\mathrm{e}^{-2t} = x + 2\left(\frac{1}{3}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{2}{3}x - \frac{1}{3}\mathrm{e}^{-2t}\right) + \mathrm{e}^{-2t}$

$\Rightarrow \frac{1}{3}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{3}\mathrm{e}^{-2t} = x + \frac{2}{3}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{4}{3}x - \frac{2}{3}\mathrm{e}^{-2t} + \mathrm{e}^{-2t}$

$\Rightarrow \frac{1}{3}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{4}{3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{3}x = -\frac{1}{3}\mathrm{e}^{-2t}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + x = -\mathrm{e}^{-2t}$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + x = 0$

Auxiliary Equation:

$m^2 - 4m + 1 = 0$

$\Rightarrow (m-2)^2 - 4 + 1 = 0$

$\Rightarrow (m-2)^2 = 3$

$\Rightarrow m = 2 + \sqrt{3}$ or $m = 2 - \sqrt{3}$

Complementary Function:

$x = A\mathrm{e}^{(2+\sqrt{3})t} + B\mathrm{e}^{(2-\sqrt{3})t}$

Try this Particular Integral:

$x = k\mathrm{e}^{-2t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2k\mathrm{e}^{-2t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 4k\mathrm{e}^{-2t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + x = -\mathrm{e}^{-2t}$

$\Rightarrow (4k\mathrm{e}^{-2t}) - 4(-2k\mathrm{e}^{-2t}) + (k\mathrm{e}^{-2t}) = -\mathrm{e}^{-2t}$

$\Rightarrow 4k\mathrm{e}^{-2t} + 8k\mathrm{e}^{-2t} + k\mathrm{e}^{-2t} = -\mathrm{e}^{-2t}$

$\Rightarrow 13k\mathrm{e}^{-2t} = -\mathrm{e}^{-2t}$

$\Rightarrow 13k = -1$

$\Rightarrow k = -\frac{1}{13}$

Particular Integral:

$x = -\frac{1}{13}\mathrm{e}^{-2t}$

General Solution:

$x = A\mathrm{e}^{(2+\sqrt{3})t} + B\mathrm{e}^{(2-\sqrt{3})t} - \frac{1}{13}\mathrm{e}^{-2t}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = A\left(2+\sqrt{3}\right)\mathrm{e}^{(2+\sqrt{3})t} + B\left(2-\sqrt{3}\right)\mathrm{e}^{(2-\sqrt{3})t} + \frac{2}{13}\mathrm{e}^{-2t}$

Then:

$y = \frac{1}{3}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{2}{3}x - \frac{1}{3}\mathrm{e}^{-2t}$

$\Rightarrow y = \frac{1}{3}\left(A\left(2+\sqrt{3}\right)\mathrm{e}^{(2+\sqrt{3})t} + B\left(2-\sqrt{3}\right)\mathrm{e}^{(2-\sqrt{3})t} + \frac{2}{13}\mathrm{e}^{-2t}\right) - \frac{2}{3}\left(A\mathrm{e}^{(2+\sqrt{3})t} + B\mathrm{e}^{(2-\sqrt{3})t} - \frac{1}{13}\mathrm{e}^{-2t}\right) - \frac{1}{3}\mathrm{e}^{-2t}$

$\Rightarrow y = \frac{1}{3}A\left(2+\sqrt{3}\right)\mathrm{e}^{(2+\sqrt{3})t} + \frac{1}{3}B\left(2-\sqrt{3}\right)\mathrm{e}^{(2-\sqrt{3})t} + \frac{2}{39}\mathrm{e}^{-2t} - \frac{2}{3}A\mathrm{e}^{(2+\sqrt{3})t} - \frac{2}{3}B\mathrm{e}^{(2-\sqrt{3})t} + \frac{2}{39}\mathrm{e}^{-2t} - \frac{1}{3}\mathrm{e}^{-2t}$

$\Rightarrow y = \frac{1}{3}\left(A\left(2+\sqrt{3}\right) - 2A\right)\mathrm{e}^{(2+\sqrt{3})t} + \frac{1}{3}\left(B\left(2-\sqrt{3}\right) - 2B\right)\mathrm{e}^{(2-\sqrt{3})t} - \frac{3}{13}\mathrm{e}^{-2t}$

$\Rightarrow y = \frac{\sqrt{3}}{3}A\mathrm{e}^{(2+\sqrt{3})t} - \frac{\sqrt{3}}{3}B\mathrm{e}^{(2-\sqrt{3})t} - \frac{3}{13}\mathrm{e}^{-2t}$

So:

$x = A\mathrm{e}^{(2+\sqrt{3})t} + B\mathrm{e}^{(2-\sqrt{3})t} - \frac{1}{13}\mathrm{e}^{-2t}$ and $y = \frac{\sqrt{3}}{3}A\mathrm{e}^{(2+\sqrt{3})t} - \frac{\sqrt{3}}{3}B\mathrm{e}^{(2-\sqrt{3})t} - \frac{3}{13}\mathrm{e}^{-2t}$

$x = A\mathrm{e}^{(2+\sqrt{3})t} + B\mathrm{e}^{(2-\sqrt{3})t} - \frac{1}{13}\mathrm{e}^{-2t}$

Substitute $t = 0$ and $x = \frac{25}{13}$:

$\frac{25}{13} = A\mathrm{e}^{0} + B\mathrm{e}^{0} - \frac{1}{13}\mathrm{e}^{0}$

$\Rightarrow A + B = 2$ (1)

$y = \frac{\sqrt{3}}{3}A\mathrm{e}^{(2+\sqrt{3})t} - \frac{\sqrt{3}}{3}B\mathrm{e}^{(2-\sqrt{3})t} - \frac{3}{13}\mathrm{e}^{-2t}$

Substitute $t = 0$ and $y = -\frac{3}{13}$:

$-\frac{3}{13} = \frac{\sqrt{3}}{3}A\mathrm{e}^{0} - \frac{\sqrt{3}}{3}B\mathrm{e}^{0} - \frac{3}{13}\mathrm{e}^{0}$ ›

$\Rightarrow \frac{\sqrt{3}}{3}A - \frac{\sqrt{3}}{3}B = 0$ (2)

Solve (1) & (2)

$A = 1$ and $B = 1$

$\therefore x = \mathrm{e}^{(2+\sqrt{3})t} + \mathrm{e}^{(2-\sqrt{3})t} - \frac{1}{13}\mathrm{e}^{-2t}$ and $y = \frac{\sqrt{3}}{3}\mathrm{e}^{(2+\sqrt{3})t} - \frac{\sqrt{3}}{3}\mathrm{e}^{(2-\sqrt{3})t} - \frac{3}{13}\mathrm{e}^{-2t}$

15)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = -3x + 4y + \mathrm{e}^{-t}$

$\Rightarrow y = \frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{4}x - \frac{1}{4}\mathrm{e}^{-t}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{4}x - \frac{1}{4}\mathrm{e}^{-t}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{3}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{4}\mathrm{e}^{-t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{2}x - 4y + \mathrm{e}^{-t}$

$\Rightarrow \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{3}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{4}\mathrm{e}^{-t} = \frac{3}{2}x - 4\left(\frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{4}x - \frac{1}{4}\mathrm{e}^{-t}\right) + \mathrm{e}^{-t}$

$\Rightarrow \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{3}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{4}\mathrm{e}^{-t} = \frac{3}{2}x - \frac{\mathrm{d}x}{\mathrm{d}t} - 3x + \mathrm{e}^{-t} + \mathrm{e}^{-t}$

$\Rightarrow \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{7}{4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{2}x = \frac{7}{4}\mathrm{e}^{-t}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7\frac{\mathrm{d}x}{\mathrm{d}t} + 6x = 7\mathrm{e}^{-t}$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7\frac{\mathrm{d}x}{\mathrm{d}t} + 6x = 0$

Auxiliary Equation:

$m^2 + 7m + 6 = 0$

$\Rightarrow (m+1)(m+6) = 0$

$\Rightarrow m = -1$ or $m = -6$

Complementary Function:

$x = A\mathrm{e}^{-t} + B\mathrm{e}^{-6t}$

Try this Particular Integral:

$x = kt\mathrm{e}^{-t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -kt\mathrm{e}^{-t} + k\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = kt\mathrm{e}^{-t} - k\mathrm{e}^{-t} - k\mathrm{e}^{-t}$

$= kt\mathrm{e}^{-t} - 2k\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7\frac{\mathrm{d}x}{\mathrm{d}t} + 6x = 7\mathrm{e}^{-t}$

$\Rightarrow (kt\mathrm{e}^{-t} - 2k\mathrm{e}^{-t}) + 7(-kt\mathrm{e}^{-t} + k\mathrm{e}^{-t}) + 6(kt\mathrm{e}^{-t}) = 7\mathrm{e}^{-t}$

$\Rightarrow (k - 7k + 6k)t\mathrm{e}^{-t} + (-2k + 7k)\mathrm{e}^{-t} = 7\mathrm{e}^{-t}$

$\Rightarrow 5k\mathrm{e}^{-t} = 7\mathrm{e}^{-t}$

$\Rightarrow 5k = 7$

$\Rightarrow k = \frac{7}{5}$

Particular Integral:

$x = \frac{7}{5}te^{-t}$

General Solution:

$x = Ae^{-t} + Be^{-6t} + \frac{7}{5}te^{-t}$

$\Rightarrow \frac{dx}{dt} = -Ae^{-t} - 6Be^{-6t} - \frac{7}{5}te^{-t} + \frac{7}{5}e^{-t}$

Then:

$y = \frac{1}{4}\frac{dx}{dt} + \frac{3}{4}x - \frac{1}{4}e^{-t}$

$\Rightarrow y = \frac{1}{4}\left(-Ae^{-t} - 6Be^{-6t} - \frac{7}{5}te^{-t} + \frac{7}{5}e^{-t}\right) + \frac{3}{4}\left(Ae^{-t} + Be^{-6t} + \frac{7}{5}te^{-t}\right) - \frac{1}{4}e^{-t}$

$\Rightarrow y = -\frac{1}{4}Ae^{-t} - \frac{3}{2}Be^{-6t} - \frac{7}{20}te^{-t} + \frac{7}{20}e^{-t} + \frac{3}{4}Ae^{-t} + \frac{3}{4}Be^{-6t} + \frac{21}{20}te^{-t} - \frac{1}{4}e^{-t}$

$\Rightarrow y = \left(-\frac{1}{4}A + \frac{7}{20} + \frac{3}{4}A - \frac{1}{4}\right)e^{-t} + \left(-\frac{3}{2}B + \frac{3}{4}B\right)e^{-6t} + \left(-\frac{7}{20} + \frac{21}{20}\right)te^{-t}$

$\Rightarrow y = \left(\frac{1}{2}A + \frac{1}{10}\right)e^{-t} - \frac{3}{4}Be^{-6t} + \frac{7}{10}te^{-t}$

So:

$x = Ae^{-t} + Be^{-6t} + \frac{7}{5}te^{-t}$ and $y = \left(\frac{1}{2}A + \frac{1}{10}\right)e^{-t} - \frac{3}{4}Be^{-6t} + \frac{7}{10}te^{-t}$

$x = Ae^{-t} + Be^{-6t} + \frac{7}{5}te^{-t}$

Substitute $t = 0$ and $x = 1$:

$1 = Ae^{0} + Be^{0} + \frac{7}{5} \times 0 \times e^{0}$

$\Rightarrow A + B = 1$ (1)

$y = \left(\frac{1}{2}A + \frac{1}{10}\right)e^{-t} - \frac{3}{4}Be^{-6t} + \frac{7}{10}te^{-t}$

Substitute $t = 0$ and $y = \frac{1}{4}$:

$\frac{1}{4} = \left(\frac{1}{2}A + \frac{1}{10}\right)e^{0} - \frac{3}{4}Be^{0} + \frac{7}{10} \times 0 \times e^{0}$

$\Rightarrow \frac{1}{4} = \frac{1}{2}A + \frac{1}{10} - \frac{3}{4}B$

$\Rightarrow \frac{1}{2}A - \frac{3}{4}B = \frac{3}{20}$ (2)

Solve (1) & (2)

$A = \frac{18}{25}$ and $B = \frac{7}{25}$

$\therefore x = \frac{18}{25}e^{-t} + \frac{7}{25}e^{-6t} + \frac{7}{5}te^{-t}$ and $y = \frac{23}{50}e^{-t} - \frac{21}{100}e^{-6t} + \frac{7}{10}te^{-t}$

16) a)

The $0.3x$ term represents the natural growth of the rabbit population.
The $-12y$ term represents the decrease in rabbits due to predation by foxes.

16) b)

The $0.02x$ term represents the increase in foxes due to the availability of rabbits as food.
The -1.5 term represents the **constant** loss of foxes each year. As it does not involve x or y, it means that foxes are lost at a fixed rate, regardless of how many foxes there are.

16) c)

If there are no foxes, then $y = 0$:
$\frac{\mathrm{d}x}{\mathrm{d}t} = 0.3x$
Hence, as $0.3x > 0$, the rabbit population would increase exponentially.

16) d)

If there are no rabbits, then $x = 0$:
$\frac{\mathrm{d}y}{\mathrm{d}t} = -1.5$
Hence the fox population would decrease steadily at 1.5 foxes per year.

16) e)

In a real population, deaths would usually depend on how many foxes there are. If there are more foxes, you would expect more deaths overall, and if there were few foxes then you would expect fewer deaths. In reality, deaths would usually be proportional to the fox population, so a term like $-1.5y$ would be more realistic.

17) a)

0.5 is the coefficient of x in the $\frac{\mathrm{d}x}{\mathrm{d}t}$ equation, and so represents the natural growth rate of the fish population.

-8 is the coefficient of y in the $\frac{\mathrm{d}x}{\mathrm{d}t}$ equation, and so represents the decrease in fish due to predation by birds.

0.02 is the coefficient of x in the $\frac{\mathrm{d}y}{\mathrm{d}t}$ equation, and so represents the increase in birds due to the availability of fish as food.

-1.6 is a constant in the $\frac{\mathrm{d}y}{\mathrm{d}t}$ equation, and so represents a fixed decrease in the bird population.

17) b)

The equilibrium point (when both populations are constant), exists when both $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$:

$0.5x - 8y = 0$ (1)

$0.02x - 1.6 = 0$ (2)

Solve (2):

$x = \frac{1.6}{0.02} = 80$

Substitute into (1):

$0.5 \times 80 - 8y = 0$

$\Rightarrow y = 5$

So the populations will remain constant if there are 80 fish and 5 birds.

18) a)

$p = 40$

$q = 4$

$r = 2$

$s = 2$

18) b)

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0.4x - 4y$

$\Rightarrow y = \frac{1}{10}x - \frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{10}x - \frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{10}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0.02x - 2$

$\Rightarrow \frac{1}{10}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0.02x - 2$

$\Rightarrow -\frac{1}{4}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}x}{\mathrm{d}t} - 0.02x = -2$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{5}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{25}x = 8$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{5}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{25}x = 0$

Auxiliary Equation:

$m^2 - \frac{2}{5}m + \frac{2}{25} = 0$

$\Rightarrow \left(m - \frac{1}{5}\right)^2 - \frac{1}{25} + \frac{2}{25} = 0$

$\Rightarrow \left(m - \frac{1}{5}\right)^2 = -\frac{1}{25}$

$\Rightarrow m = \frac{1}{5} \pm \frac{1}{5}\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$

Try this Particular Integral:

$x = k$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{2}{5}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{25}x = 8$

$\Rightarrow (0) - \frac{2}{5}(0) + \frac{2}{25}(k) = 8$

$\Rightarrow \frac{2}{25}k = 8$

$\Rightarrow k = 100$

Particular Integral:

$x = 100$

General Solution:

$x = \mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right) + 100$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{\frac{1}{5}t}\left(-\frac{1}{5}A\sin\frac{1}{5}t + \frac{1}{5}B\cos\frac{1}{5}t\right) + \frac{1}{5}\mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$

Then:

$y = \frac{1}{10}x - \frac{1}{4}\frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow y = \frac{1}{10}\left(\mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right) + 100\right) - \frac{1}{4}\left(\mathrm{e}^{\frac{1}{5}t}\left(-\frac{1}{5}A\sin\frac{1}{5}t + \frac{1}{5}B\cos\frac{1}{5}t\right) + \frac{1}{5}\mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)\right)$

$\Rightarrow y = \mathrm{e}^{\frac{1}{5}t}\left(\frac{1}{10}A\cos\frac{1}{5}t + \frac{1}{10}B\sin\frac{1}{5}t\right) + 10 + \mathrm{e}^{\frac{1}{5}t}\left(\frac{1}{20}A\sin\frac{1}{5}t - \frac{1}{20}B\cos\frac{1}{5}t\right) + \mathrm{e}^{\frac{1}{5}t}\left(-\frac{1}{20}A\cos\frac{1}{5}t - \frac{1}{20}B\sin\frac{1}{5}t\right)$

$\Rightarrow y = \mathrm{e}^{\frac{1}{5}t}\left(\frac{1}{10}A\cos\frac{1}{5}t + \frac{1}{10}B\sin\frac{1}{5}t + \frac{1}{20}A\sin\frac{1}{5}t - \frac{1}{20}B\cos\frac{1}{5}t - \frac{1}{20}A\cos\frac{1}{5}t - \frac{1}{20}B\sin\frac{1}{5}t\right) + 10$

$\Rightarrow y = \mathrm{e}^{\frac{1}{5}t}\left(\left(\frac{1}{10}A - \frac{1}{20}B - \frac{1}{20}A\right)\cos\frac{1}{5}t + \left(\frac{1}{10}B + \frac{1}{20}A - \frac{1}{20}B\right)\sin\frac{1}{5}t\right) + 10$

$\Rightarrow y = \mathrm{e}^{\frac{1}{5}t}\left(\left(\frac{1}{20}A - \frac{1}{20}B\right)\cos\frac{1}{5}t + \left(\frac{1}{20}A + \frac{1}{20}B\right)\sin\frac{1}{5}t\right) + 10$

So:

$x = \mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right) + 100$ and $y = \mathrm{e}^{\frac{1}{5}t}\left(\left(\frac{1}{20}A - \frac{1}{20}B\right)\cos\frac{1}{5}t + \left(\frac{1}{20}A + \frac{1}{20}B\right)\sin\frac{1}{5}t\right) + 10$

$x = \mathrm{e}^{\frac{1}{5}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right) + 100$

Substitute $t = 0$ and $x = 1200$:

$1200 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + 100$

$\Rightarrow 1200 = A + 100$

$\Rightarrow A = 1100$

$y = \mathrm{e}^{\frac{1}{5}t}\left(\left(55 - \frac{1}{20}B\right)\cos\frac{1}{5}t + \left(55 + \frac{1}{20}B\right)\sin\frac{1}{5}t\right) + 10$

Substitute $t = 0$ and $y = 25$:

$25 = \mathrm{e}^0\left(\left(55 - \frac{1}{20}B\right)\cos 0 + \left(55 + \frac{1}{20}B\right)\sin 0\right) + 10$

$\Rightarrow 25 = 55 - \frac{1}{20}B + 10$

$\Rightarrow B = 800$

$\therefore x = \mathrm{e}^{\frac{1}{5}t}\left(1100\cos\frac{1}{5}t + 800\sin\frac{1}{5}t\right) + 100$ and $y = \mathrm{e}^{\frac{1}{5}t}\left(15\cos\frac{1}{5}t + 95\sin\frac{1}{5}t\right) + 10$

18) c)

$x|_{t=4} = \mathrm{e}^{\frac{4}{5}}\left(1100\cos\frac{4}{5} + 800\sin\frac{4}{5}\right) + 100$

$= 3082.808\ldots$

≈ 3083

The initial population of insects is 1200, so the population of insects has increased. Therefore, the insect population is not being successfully controlled.

19) a)

$A = -0.05$

$B = -2$

$C = 0.02$

$D = -0.05$

19) b)

The owl death term being proportional to y means that owl deaths depend on the size of the owl population: the larger the owl population, the more owls die each year.

19) c)

$\frac{\mathrm{d}x}{\mathrm{d}t} = -0.05x - 2y$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0.02x - 0.05y$

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = -0.05x - 2y$

$\Rightarrow y = -\frac{1}{40}x - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{1}{40}x - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{40}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0.02x - 0.05y$

$\Rightarrow -\frac{1}{40}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0.02x - 0.05\left(-\frac{1}{40}x - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t}\right)$

$\Rightarrow -\frac{1}{40}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{1}{50}x + \frac{1}{800}x + \frac{1}{40}\frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow -\frac{1}{2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - \frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{17}{800}x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{17}{400}x = 0$

Auxiliary Equation:

$m^2 + \frac{1}{10}m + \frac{17}{400} = 0$

$\Rightarrow \left(m + \frac{1}{20}\right)^2 - \frac{1}{400} + \frac{17}{400} = 0$

$\Rightarrow \left(m + \frac{1}{20}\right)^2 = -\frac{1}{25}$

$\Rightarrow m = -\frac{1}{20} \pm \frac{1}{5}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{5}A\sin\frac{1}{5}t + \frac{1}{5}B\cos\frac{1}{5}t\right) - \frac{1}{20}\mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$

Then:

$y = -\frac{1}{40}x - \frac{1}{2}\frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow y = -\frac{1}{40}\left(\mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)\right) - \frac{1}{2}\left(\mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{5}A\sin\frac{1}{5}t + \frac{1}{5}B\cos\frac{1}{5}t\right) - \frac{1}{20}\mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)\right)$

$\Rightarrow y = \mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{40}A\cos\frac{1}{5}t - \frac{1}{40}B\sin\frac{1}{5}t\right) + \mathrm{e}^{-\frac{1}{20}t}\left(\frac{1}{10}A\sin\frac{1}{5}t - \frac{1}{10}B\cos\frac{1}{5}t\right) +$
$\mathrm{e}^{-\frac{1}{20}t}\left(\frac{1}{40}A\cos\frac{1}{5}t + \frac{1}{40}B\sin\frac{1}{5}t\right)$

$\Rightarrow y = \mathrm{e}^{-\frac{1}{20}t}\left(\left(-\frac{1}{40}A - \frac{1}{10}B + \frac{1}{40}A\right)\cos\frac{1}{5}t + \left(-\frac{1}{40}B + \frac{1}{10}A + \frac{1}{40}B\right)\sin\frac{1}{5}t\right)$

$\Rightarrow y = \mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{10}B\cos\frac{1}{5}t + \frac{1}{10}A\sin\frac{1}{5}t\right)$

So:

$x = \mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$ and $y = \mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{10}B\cos\frac{1}{5}t + \frac{1}{10}A\sin\frac{1}{5}t\right)$

$x = \mathrm{e}^{-\frac{1}{20}t}\left(A\cos\frac{1}{5}t + B\sin\frac{1}{5}t\right)$

Substitute $t = 0$ and $x = 300$:

$300 = \mathrm{e}^{0}(A\cos 0 + B\sin 0)$

$\Rightarrow A = 300$

$y = \mathrm{e}^{-\frac{1}{20}t}\left(-\frac{1}{10}B\cos\frac{1}{5}t + 30\sin\frac{1}{5}t\right)$

Substitute $t = 0$ and $y = 20$:

$20 = \mathrm{e}^{0}\left(-\frac{1}{10}B\cos 0 + 30\sin 0\right)$

$\Rightarrow 20 = -\frac{1}{10}B$

$\Rightarrow B = -200$

$\therefore x = \mathrm{e}^{-\frac{1}{20}t}\left(300\cos\frac{1}{5}t - 200\sin\frac{1}{5}t\right)$ and $y = \mathrm{e}^{-\frac{1}{20}t}\left(20\cos\frac{1}{5}t + 30\sin\frac{1}{5}t\right)$

19) d)

The vole population is zero when $x = 0$:

$e^{-\frac{1}{20}t}\left(300\cos\frac{1}{5}t - 200\sin\frac{1}{5}t\right) = 0$

$e^{-\frac{1}{20}t} \neq 0$ so:

$\Rightarrow 300\cos\frac{1}{5}t - 200\sin\frac{1}{5}t = 0$

$\Rightarrow 3\cos\frac{1}{5}t = 2\sin\frac{1}{5}t$

$\Rightarrow \frac{\sin\frac{1}{5}t}{\cos\frac{1}{5}t} = \frac{3}{2}$

$\Rightarrow \tan\frac{1}{5}t = \frac{3}{2}$

$\Rightarrow \frac{1}{5}t = \arctan\frac{3}{2}$

$\Rightarrow t = 5\arctan\frac{3}{2} = 4.91$ (3 SF)

So the vole population becomes zero after approximately 4.91 years. After this time, the equation for $x(t)$ would give negative values for the vole population, and so the model stops making sense.

20) a)

$\text{Concentration} = \frac{\text{amount of salt}}{\text{volume of liquid}}$

$= \frac{x}{500}$ grams per litre at time t

20) b)

Salt leaves with the outgoing liquid. The outflow is 8 litres per minute, and the concentration in the tank is $\frac{x}{500}$ grams per litre. So:

rate out $= 8 \times \frac{x}{500} = \frac{2}{125}x$ grams per minute

20) c)

$\frac{dx}{dt} = \text{rate in} - \text{rate out}$

The incoming solution has concentration 4 g/l and flows in at 8 l/min so

rate in $= 4 \times 8 = 32$ g/min

$\therefore \frac{dx}{dt} = 32 - \frac{2}{125}x$

20) d)

Liquid from tank 1 flows into tank 2 at 8 l/min. The concentration in tank 1 is $\frac{x}{500}$ g/l, so the rate at which salt enters tank 2 from tank 1 is:

$8 \times \frac{x}{500} = \frac{2}{125}x$

Extra salt solution flows directly into tank 2 with concentration 10 g/l and flow rate 2 l/min. So this contributes salt at rate:

$10 \times 2 = 20$ g/min

Tank 2's concentration is $\frac{y}{500}$ g/l and liquid leaves tank 2 at 10 l/m, so the rate at which salt leaves is:

$10 \times \frac{y}{500} = \frac{1}{50}y$

$\therefore \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{2}{125}x + 20 - \frac{1}{50}y$

21) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

No liquid flows into tank A from outside, so:

$\text{rate in} = 0$

Liquid flows from A to B at 12 litres per minute. The concentration of salt in tank A is $\frac{x}{400}$ grams per litre. So:

$\text{rate out} = 12 \times \frac{x}{400} = \frac{3}{100}x$

$\therefore \frac{\mathrm{d}x}{\mathrm{d}t} = 0 - \frac{3}{100}x$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{3}{100}x$

21) b)

$\frac{\mathrm{d}y}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

Liquid enters tank B from tank A at 12 litres per minute. The concentration of salt in tank A is $\frac{x}{400}$ grams per litre. So:

$\text{rate in} = 12 \times \frac{x}{400} = \frac{3}{100}x$

Liquid leaves tank B at 12 litres per minute. The concentration of salt in tank B is $\frac{y}{400}$ grams per litre. So:

$\text{rate out} = 12 \times \frac{y}{400} = \frac{3}{100}y$

$\therefore \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{100}x - \frac{3}{100}y$

21) c)

Use Separation of Variables to solve $\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{3}{100}x$

$\Rightarrow \int \frac{1}{x}\mathrm{d}x = \int -\frac{3}{100}\mathrm{d}t$

$\Rightarrow \ln|x| = -\frac{3}{100}t + c$

$\Rightarrow x = A\mathrm{e}^{-\frac{3}{100}t}$

Substitute $t = 0$ and $x = 60$:

$60 = A\mathrm{e}^{0}$

$\Rightarrow A = 60$

So $x = 60\mathrm{e}^{-\frac{3}{100}t}$

Substitute into $\frac{\mathrm{d}y}{\mathrm{d}t}$:

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{100}\left(60\mathrm{e}^{-\frac{3}{100}t}\right) - \frac{3}{100}y$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{9}{5}\mathrm{e}^{-\frac{3}{100}t} - \frac{3}{100}y$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} + \frac{3}{100}y = \frac{9}{5}\mathrm{e}^{-\frac{3}{100}t}$

Use an Integrating Factor to solve $\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{3}{100}y = \frac{9}{5}\mathrm{e}^{-\frac{3}{100}t}$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{100}\mathrm{d}t}$

$= \mathrm{e}^{\frac{3}{100}t}$

Let Integrating Factor $= \mathrm{e}^{\frac{3}{100}t}$

$\mathrm{e}^{\frac{3}{100}t}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{3}{100}\mathrm{e}^{\frac{3}{100}t}y = \frac{9}{5}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\frac{3}{100}t}y\right) = \frac{9}{5}$

$\Rightarrow \mathrm{e}^{\frac{3}{100}t}y = \frac{9}{5}t + c$

$\Rightarrow y = \frac{9}{5}t\mathrm{e}^{-\frac{3}{100}t} + c\mathrm{e}^{-\frac{3}{100}t}$

Substitute $t = 0$ and $y = 0$:

$0 = \frac{9}{5} \times 0 \times \mathrm{e}^{0} + c\mathrm{e}^{0}$

$\Rightarrow c = 0$

So:

$x = 60\mathrm{e}^{-\frac{3}{100}t}$ and $y = \frac{9}{5}t\mathrm{e}^{-\frac{3}{100}t}$

22) a)

Tank Q receives liquid from P at 15 litres per minute, and loses liquid back to P at 7 litres per minute. So without any extra outflow, the volume in Q would increase at $15 - 7 = 8$ litres per minute. But as the volume of liquid in Q stays constant, liquid must leave Q at 8 litres per minute.

22) b)

Concentration in P is $\frac{x}{700}$

Concentration in Q is $\frac{y}{700}$

Liquid enters P from Q at 7 l/min, so salt enters at rate

$7 \times \frac{y}{700} = \frac{1}{100}y$

Liquid leaves P for Q at 15 l/min, so salt leaves at rate

$15 \times \frac{x}{700} = \frac{3}{140}x$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{100}y - \frac{3}{140}x$

Liquid enters Q from P at 15 l/min, so salt enters at rate

$15 \times \frac{x}{700} = \frac{3}{140}x$

Liquid leaves Q for P at 7 l/min, so salt leaves at rate

$7 \times \frac{y}{700} = \frac{1}{100}y$

Liquid leaves Q to the outside at 8 l/min, so salt leaves at rate

$8 \times \frac{y}{700} = \frac{2}{175}y$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{140}x - \left(\frac{1}{100}y + \frac{2}{175}y\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{140}x - \frac{3}{140}y$

So:

$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{100}y - \frac{3}{140}x$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{140}x - \frac{3}{140}y$

23) a)

Consider tank M. There are two inflows:

External salt solution enters at 7 l/min with concentration c g/l. So salt enters at rate:

$7 \times c = 7c$

Liquid flows from N to M at 9 l/min. Concentration in tank N is $\frac{y}{900}$ g/l. So salt enters at rate:

$9 \times \frac{y}{900} = \frac{1}{100}y$

Liquid flows from M to N at 16 l/min. Concentration in tank M is $\frac{x}{900}$ g/l. So salt leaves at rate:

$16 \times \frac{x}{900} = \frac{4}{225}x$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \left(7c + \frac{1}{100}y\right) - \frac{4}{225}x$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = 7c - \frac{4}{225}x + \frac{1}{100}y$

23) b)

Consider tank N.

Liquid flows from M to N at 16 l/min. Concentration in tank M is $\frac{x}{900}$ g/l. So salt enters at rate:

$16 \times \frac{x}{900} = \frac{4}{225}x$

Liquid flows from N to M at 9 l/min. Concentration in tank N is $\frac{y}{900}$ g/l. So salt leaves at rate:

$9 \times \frac{y}{900} = \frac{1}{100}y$

Liquid flows from to the outside at $(16 - 9) = 7$ l/min. Concentration in tank N is $\frac{y}{900}$ g/l. So salt leaves at rate:

$7 \times \frac{y}{900} = \frac{7}{900}y$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{4}{225}x - \left(\frac{1}{100}y + \frac{7}{900}y\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{4}{225}x - \frac{4}{225}y$

23) c)

The steady state is reached when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$

$\frac{dx}{dt} = 7c - \frac{4}{225}x + \frac{1}{100}y$

Substitute $\frac{dx}{dt} = 0$, $x = 450$ and $y = 450$:

$0 = 7c - \frac{4}{225} \times 450 + \frac{1}{100} \times 450$

$\Rightarrow c = \frac{1}{2}$

23) d)

$\frac{dx}{dt} = \frac{7}{2} - \frac{4}{225}x + \frac{1}{100}y$

$\frac{dy}{dt} = \frac{4}{225}x - \frac{4}{225}y$

Rearrange the first equation:

$\frac{dx}{dt} = \frac{7}{2} - \frac{4}{225}x + \frac{1}{100}y$

$\Rightarrow y = 100\frac{dx}{dt} + \frac{16}{9}x - 350$

$\frac{d}{dt}(y) = \frac{d}{dt}\left(100\frac{dx}{dt} + \frac{16}{9}x - 350\right)$

$\Rightarrow \frac{dy}{dt} = 100\frac{d^2x}{dt^2} + \frac{16}{9}\frac{dx}{dt}$

Hence:

$\frac{dy}{dt} = \frac{4}{225}x - \frac{4}{225}y$

$\Rightarrow 100\frac{d^2x}{dt^2} + \frac{16}{9}\frac{dx}{dt} = \frac{4}{225}x - \frac{4}{225}\left(100\frac{dx}{dt} + \frac{16}{9}x - 350\right)$

$\Rightarrow 100\frac{d^2x}{dt^2} + \frac{16}{9}\frac{dx}{dt} = \frac{4}{225}x - \frac{16}{9}\frac{dx}{dt} - \frac{64}{2025}x + \frac{56}{9}$

$\Rightarrow 100\frac{d^2x}{dt^2} + \frac{32}{9}\frac{dx}{dt} + \frac{28}{2025}x = \frac{56}{9}$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{8}{225}\frac{dx}{dt} + \frac{7}{50625}x = \frac{14}{225}$

Homogeneous Equation:

$\frac{d^2x}{dt^2} + \frac{8}{225}\frac{dx}{dt} + \frac{7}{50625}x = 0$

Auxiliary Equation:

$m^2 + \frac{8}{225}m + \frac{7}{50625} = 0$

$\Rightarrow \left(m + \frac{4}{225}\right)^2 - \frac{16}{50625} + \frac{7}{50625} = 0$

$\Rightarrow \left(m + \frac{4}{225}\right)^2 = \frac{1}{5625}$

$\Rightarrow m = -\frac{4}{225} \pm \frac{1}{75}$

$\Rightarrow m = -\frac{1}{225}$ or $m = -\frac{7}{225}$

Complementary Function:

$x = Ae^{-\frac{1}{225}t} + Be^{-\frac{7}{225}t}$

Try this Particular Integral:

$x = k$

$\frac{dx}{dt} = 0$

$\frac{d^2x}{dt^2} = 0$

$\frac{d^2x}{dt^2} + \frac{8}{225}\frac{dx}{dt} + \frac{7}{50625}x = \frac{14}{225}$

$\Rightarrow (0) + \frac{8}{225}(0) + \frac{7}{50625}(k) = \frac{14}{225}$

$\Rightarrow \frac{7}{50625}k = \frac{14}{225}$

$\Rightarrow k = 450$

Particular Integral:

$x = 450$

General Solution:

$x = Ae^{-\frac{1}{225}t} + Be^{-\frac{7}{225}t} + 450$

$\Rightarrow \frac{dx}{dt} = -\frac{1}{225}Ae^{-\frac{1}{225}t} - \frac{7}{225}Be^{-\frac{7}{225}t}$

Then:

$y = 100\frac{dx}{dt} + \frac{16}{9}x - 350$

$\Rightarrow y = 100\left(-\frac{1}{225}Ae^{-\frac{1}{225}t} - \frac{7}{225}Be^{-\frac{7}{225}t}\right) + \frac{16}{9}\left(Ae^{-\frac{1}{225}t} + Be^{-\frac{7}{225}t} + 450\right) - 350$

$\Rightarrow y = -\frac{4}{9}Ae^{-\frac{1}{225}t} - \frac{28}{9}Be^{-\frac{7}{225}t} + \frac{16}{9}Ae^{-\frac{1}{225}t} + \frac{16}{9}Be^{-\frac{7}{225}t} + 800 - 350$

$\Rightarrow y = \frac{12}{9}Ae^{-\frac{1}{225}t} - \frac{12}{9}Be^{-\frac{7}{225}t} + 450$

So:

$x = Ae^{-\frac{1}{225}t} + Be^{-\frac{7}{225}t} + 450$ and $y = \frac{12}{9}Ae^{-\frac{1}{225}t} - \frac{12}{9}Be^{-\frac{7}{225}t} + 450$

$x = Ae^{-\frac{1}{225}t} + Be^{-\frac{7}{225}t} + 450$

Substitute $t = 0$ and $x = 500$:

$500 = Ae^0 + Be^0 + 450$

$\Rightarrow A + B = 50$ (1)

$y = \frac{12}{9}Ae^{-\frac{1}{225}t} - \frac{12}{9}Be^{-\frac{7}{225}t} + 450$

Substitute $t = 0$ and $y = 400$:

$400 = \frac{12}{9}A\mathrm{e}^{0} - \frac{12}{9}B\mathrm{e}^{0} + 450$

$\Rightarrow -50 = \frac{12}{9}A - \frac{12}{9}B$

$\Rightarrow A - B = -\frac{75}{2}$ (2)

Solve (1) & (2):

$A = \frac{25}{4}$ and $B = \frac{175}{4}$

$\therefore x = \frac{25}{4}\mathrm{e}^{-\frac{1}{225}t} + \frac{175}{4}\mathrm{e}^{-\frac{7}{225}t} + 450$ and $y = \frac{25}{3}\mathrm{e}^{-\frac{1}{225}t} - \frac{175}{3}\mathrm{e}^{-\frac{7}{225}t} + 450$

24) a)

Since the volume in C stays constant, the total inflow into C must equal the total outlow from C. The outflow from C is 25 l/min, and the flow from D to C is 9 l/min. So if the external inflow into C is r l/min, then $r + 9 = 25 \Rightarrow r = 16$

Hence the rate at which liquid flows into container C is 16 l/min

24) b)

Consider container C. There are two inflows:

External salt solution enters at 16 l/min with concentration p g/l. So salt enters at rate:

$16 \times p = 16p$

Liquid flows from D to C at 9 l/min. Concentration in container D is $\frac{y}{1000}$ g/l. So salt enters at rate:

$9 \times \frac{y}{1000} = \frac{9}{1000}y = 0.009y$

Liquid flows from C to D at 25 l/min. Concentration in tank C is $\frac{x}{1000}$ g/l. So salt leaves at rate:

$25 \times \frac{x}{1000} = \frac{1}{40}x = 0.025x$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = (16p + 0.009y) - 0.025x$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = 16p - 0.025x + 0.009y$

Consider container D. There are two outflows:

Liquid flows from C to D at 25 l/min. Concentration in container C is $\frac{x}{1000}$ g/l. So salt enters at rate:

$25 \times \frac{x}{1000} = \frac{1}{40}x = 0.025x$

Liquid flows from D to C at 15 l/min. Concentration in container D is $\frac{y}{1000}$ g/l. So salt leaves at rate:

$15 \times \frac{y}{1000} = \frac{3}{200}y = 0.015y$

Liquid flows through the outlet $(25 - 15) = 10$ l/min with concentration $\frac{y}{1000}$ g/l (as the volume remains constant). So salt leaves at rate:

$10 \times \frac{y}{1000} = \frac{1}{100}y = 0.01y$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \text{rate in} - \text{rate out}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = 0.025x - (0.015y + 0.01y)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = 0.025x - 0.025y$

24) c)

$p = 4$ so:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 64 - 0.025x + 0.009y$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0.025x - 0.025y$

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 64 - 0.025x + 0.009y$

$\Rightarrow y = \frac{1000}{9}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{25}{9}x - \frac{64000}{9}$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1000}{9}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{25}{9}x - \frac{64000}{9}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1000}{9}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{25}{9}\frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0.025x - 0.025y$

$\Rightarrow \frac{1000}{9}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{25}{9}\frac{\mathrm{d}x}{\mathrm{d}t} = 0.025x - 0.025\left(\frac{1000}{9}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{25}{9}x - \frac{64000}{9}\right)$

$\Rightarrow \frac{1000}{9}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{25}{9}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{40}x - \frac{25}{9}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{5}{72}x + \frac{1600}{9}$

$\Rightarrow \frac{1000}{9}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{50}{9}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2}{45}x = \frac{1600}{9}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{2500}x = \frac{8}{5}$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{2500}x = 0$

Auxiliary Equation:

$m^2 + \frac{1}{20}m + \frac{1}{2500} = 0$

$\Rightarrow \left(m + \frac{1}{40}\right)^2 - \frac{1}{1600} + \frac{1}{2500} = 0$

$\Rightarrow \left(m + \frac{1}{40}\right)^2 = \frac{9}{40000}$

$\Rightarrow m = -\frac{1}{40} \pm \frac{3}{200}$

$\Rightarrow m = -\frac{1}{100}$ or $m = -\frac{1}{25}$

Complementary Function:

$x = A\mathrm{e}^{-\frac{1}{100}t} + B\mathrm{e}^{-\frac{1}{25}t}$

Try this Particular Integral:

$x = k$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{2500}x = \frac{8}{5}$

$\Rightarrow (0) + \frac{1}{20}(0) + \frac{1}{2500}(k) = \frac{8}{5}$

$\Rightarrow \frac{1}{2500}k = \frac{8}{5}$

$\Rightarrow k = 4000$

Particular Integral:

$x = 4000$

General Solution:

$x = A\mathrm{e}^{-\frac{1}{100}t} + B\mathrm{e}^{-\frac{1}{25}t} + 4000$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{100}A\mathrm{e}^{-\frac{1}{100}t} - \frac{1}{25}B\mathrm{e}^{-\frac{1}{25}t}$

Then: $y = \frac{1000}{9}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{25}{9}x - \frac{64000}{9}$

$\Rightarrow y = \frac{1000}{9}\left(-\frac{1}{100}A\mathrm{e}^{-\frac{1}{100}t} - \frac{1}{25}B\mathrm{e}^{-\frac{1}{25}t}\right) + \frac{25}{9}\left(A\mathrm{e}^{-\frac{1}{100}t} + B\mathrm{e}^{-\frac{1}{25}t} + 4000\right) - \frac{64000}{9}$

$\Rightarrow y = -\frac{10}{9}A\mathrm{e}^{-\frac{1}{100}t} - \frac{40}{9}B\mathrm{e}^{-\frac{1}{25}t} + \frac{25}{9}A\mathrm{e}^{-\frac{1}{100}t} + \frac{25}{9}B\mathrm{e}^{-\frac{1}{25}t} + \frac{100000}{9} - \frac{64000}{9}$

$\Rightarrow y = \frac{5}{3}A\mathrm{e}^{-\frac{1}{100}t} - \frac{5}{3}B\mathrm{e}^{-\frac{1}{25}t} + 4000$

So: $x = A\mathrm{e}^{-\frac{1}{100}t} + B\mathrm{e}^{-\frac{1}{25}t} + 4000$ and $y = \frac{15}{9}A\mathrm{e}^{-\frac{1}{100}t} - \frac{5}{3}B\mathrm{e}^{-\frac{1}{25}t} + 4000$

$x = A\mathrm{e}^{-\frac{1}{100}t} + B\mathrm{e}^{-\frac{1}{25}t} + 4000$

Substitute $t = 0$ and $x = 0$:

$0 = A\mathrm{e}^0 + B\mathrm{e}^0 + 4000$

$\Rightarrow A + B = -4000$ (1)

$y = \frac{5}{3}A\mathrm{e}^{-\frac{1}{100}t} - \frac{5}{3}B\mathrm{e}^{-\frac{1}{25}t} + 4000$

Substitute $t = 0$ and $y = 0$:

$0 = \frac{5}{3}A\mathrm{e}^0 - \frac{5}{3}B\mathrm{e}^0 + 4000$

$\Rightarrow -4000 = \frac{5}{3}A - \frac{5}{3}B$

$\Rightarrow A - B = -2400$ (2)

Solve (1) & (2):

$A = -3200$ and $B = -800$

$\therefore x = -3200\mathrm{e}^{-\frac{1}{100}t} - 800\mathrm{e}^{-\frac{1}{25}t} + 4000$ and $y = -\frac{16000}{3}\mathrm{e}^{-\frac{1}{100}t} + \frac{4000}{3}\mathrm{e}^{-\frac{1}{25}t} + 4000$

24) d)

As $t \to \infty$, $e^{-\frac{1}{100}t} \to 0$ and $e^{-\frac{1}{25}t} \to 0$

So $x \to 4000$ and $y \to 4000$

Since each container contains 1000 litres, the limiting concentration in each is:

$\frac{4000}{1000} = 4$ g/l

25) a)

$\frac{da}{dt} = -3a + 4b + 6$

$\frac{db}{dt} = -2a + 2b + 10$

Rearrange the first equation:

$\frac{da}{dt} = -3a + 4b + 6$

$\Rightarrow b = \frac{1}{4}\frac{da}{dt} + \frac{3}{4}a - \frac{3}{2}$

$\frac{d}{dt}(b) = \frac{d}{dt}\left(\frac{1}{4}\frac{da}{dt} + \frac{3}{4}a - \frac{3}{2}\right)$

$\Rightarrow \frac{db}{dt} = \frac{1}{4}\frac{d^2a}{dt^2} + \frac{3}{4}\frac{da}{dt}$

Hence:

$\frac{db}{dt} = -2a + 2b + 10$

$\Rightarrow \frac{1}{4}\frac{d^2a}{dt^2} + \frac{3}{4}\frac{da}{dt} = -2a + 2\left(\frac{1}{4}\frac{da}{dt} + \frac{3}{4}a - \frac{3}{2}\right) + 10$

$\Rightarrow \frac{1}{4}\frac{d^2a}{dt^2} + \frac{3}{4}\frac{da}{dt} = -2a + \frac{1}{2}\frac{da}{dt} + \frac{3}{2}a - 3 + 10$

$\Rightarrow \frac{1}{4}\frac{d^2a}{dt^2} + \frac{1}{4}\frac{da}{dt} + \frac{1}{2}a = 7$

$\Rightarrow \frac{d^2a}{dt^2} + \frac{da}{dt} + 2a = 28$

25) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2a}{\mathrm{d}t^2} + \frac{\mathrm{d}a}{\mathrm{d}t} + 2a = 0$

Auxiliary Equation:

$m^2 + m + 2 = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 - \frac{1}{4} + 2 = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = -\frac{7}{4}$

$\Rightarrow m = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}$

Complementary Function:

$a = \mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right)$

Try this Particular Integral:

$a = k$

$\frac{\mathrm{d}a}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2a}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2a}{\mathrm{d}t^2} + \frac{\mathrm{d}a}{\mathrm{d}t} + 2a = 28$

$\Rightarrow (0) + (0) + 2(k) = 28$

$\Rightarrow 2k = 28$

$\Rightarrow k = 14$

Particular Integral:

$a = 14$

General Solution:

$a = \mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 14$

25) c)

$$a = \mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 14$$

$$\Rightarrow \frac{\mathrm{d}a}{\mathrm{d}t} = \mathrm{e}^{-\frac{1}{2}t}\left(-\frac{\sqrt{7}}{2}A\sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{\sqrt{7}}{2}B\cos\left(\frac{\sqrt{7}}{2}t\right)\right) - \frac{1}{2}\mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right)$$

Then:

$$b = \frac{1}{4}\frac{\mathrm{d}a}{\mathrm{d}t} + \frac{3}{4}a - \frac{3}{2}$$

$$\Rightarrow b = \frac{1}{4}\left(\mathrm{e}^{-\frac{1}{2}t}\left(-\frac{\sqrt{7}}{2}A\sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{\sqrt{7}}{2}B\cos\left(\frac{\sqrt{7}}{2}t\right)\right) - \frac{1}{2}\mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right)\right) + \frac{3}{4}\left(\mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 14\right) - \frac{3}{2}$$

$$\Rightarrow b = \mathrm{e}^{-\frac{1}{2}t}\left(-\frac{\sqrt{7}}{8}A\sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{\sqrt{7}}{8}B\cos\left(\frac{\sqrt{7}}{2}t\right)\right) + \mathrm{e}^{-\frac{1}{2}t}\left(-\frac{1}{8}A\cos\left(\frac{\sqrt{7}}{2}t\right) - \frac{1}{8}B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + \mathrm{e}^{-\frac{1}{2}t}\left(\frac{3}{4}A\cos\left(\frac{\sqrt{7}}{2}t\right) + \frac{3}{4}B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + \frac{21}{2} - \frac{3}{2}$$

$$\Rightarrow b = \mathrm{e}^{-\frac{1}{2}t}\left(-\frac{\sqrt{7}}{8}A\sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{\sqrt{7}}{8}B\cos\left(\frac{\sqrt{7}}{2}t\right) - \frac{1}{8}A\cos\left(\frac{\sqrt{7}}{2}t\right) - \frac{1}{8}B\sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{3}{4}A\cos\left(\frac{\sqrt{7}}{2}t\right) + \frac{3}{4}B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 9$$

$$\Rightarrow b = \mathrm{e}^{-\frac{1}{2}t}\left(\left(\frac{\sqrt{7}}{8}B - \frac{1}{8}A + \frac{3}{4}A\right)\cos\left(\frac{\sqrt{7}}{2}t\right) + \left(-\frac{\sqrt{7}}{8}A - \frac{1}{8}B + \frac{3}{4}B\right)\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 9$$

$$\Rightarrow b = \mathrm{e}^{-\frac{1}{2}t}\left(\left(\frac{5}{8}A + \frac{\sqrt{7}}{8}B\right)\cos\left(\frac{\sqrt{7}}{2}t\right) + \left(-\frac{\sqrt{7}}{8}A + \frac{5}{8}B\right)\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 9$$

25) d)

$a = \mathrm{e}^{-\frac{1}{2}t}\left(A\cos\left(\frac{\sqrt{7}}{2}t\right) + B\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 14$

Substitute $t = 0$ and $a = 4$:

$4 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + 14$

$\Rightarrow 4 = A + 14$

$\Rightarrow A = -10$

$b = \mathrm{e}^{-\frac{1}{2}t}\left(\left(-\frac{25}{4} + \frac{\sqrt{7}}{8}B\right)\cos\left(\frac{\sqrt{7}}{2}t\right) + \left(\frac{5\sqrt{7}}{4} + \frac{5}{8}B\right)\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 9$

Substitute $t = 0$ and $b = 1$:

$1 = \mathrm{e}^{0}\left(\left(-\frac{25}{4} + \frac{\sqrt{7}}{8}B\right)\cos 0 + \left(\frac{5\sqrt{7}}{4} + \frac{5}{8}B\right)\sin 0\right) + 9$

$\Rightarrow 1 = -\frac{25}{4} + \frac{\sqrt{7}}{8}B + 9$

$\Rightarrow B = -2\sqrt{7}$

$\therefore a = \mathrm{e}^{-\frac{1}{2}t}\left(-10\cos\left(\frac{\sqrt{7}}{2}t\right) - 2\sqrt{7}\sin\left(\frac{\sqrt{7}}{2}t\right)\right) + 14$ and $b = -8\mathrm{e}^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{7}}{2}t\right) + 9$

25) e)

As $t \to \infty$, $\mathrm{e}^{-\frac{1}{2}t} \to 0$ so $a \to 14$ and $b \to 9$

So the steady state is $a = 14$ and $b = 9$, and 14 and 9 are the limiting values of a and b respectively.

25) f)

$a|_{t=6} = \mathrm{e}^{-\frac{1}{2}\times 6}\left(-10\cos\left(\frac{\sqrt{7}}{2}\times 6\right) - 2\sqrt{7}\sin\left(\frac{\sqrt{7}}{2}\times 6\right)\right) + 14$

$= \mathrm{e}^{-3}\left(-10\cos\left(3\sqrt{7}\right) - 2\sqrt{7}\sin\left(3\sqrt{7}\right)\right) + 14$

$= 13.7788\ldots$

$b|_{t=6} = -8\mathrm{e}^{-\frac{1}{2}\times 6}\cos\left(\frac{\sqrt{7}}{2}\times 6\right) + 9$

$= -8\mathrm{e}^{-3}\cos\left(3\sqrt{7}\right) + 9$

$= 9.0331\ldots$

So after 6 minutes, the masses are very close to their limiting values, so the chemist's claim is reasonably plausible in a practical sense.

However, although $\mathrm{e}^{-\frac{1}{2}t}$ tends to 0, the sine and cosine terms means the solution approaches $a = 14$ and $b = 9$ in an oscillatory way, spiralling inwards rather than moving straight towards it. This means that after $t = 6$, the system may move slightly further away from equilibrium before coming closer again.

When $t = 7$:

$a|_{t=7} = \mathrm{e}^{-\frac{1}{2}\times 7}\left(-10\cos\left(\frac{\sqrt{7}}{2}\times 7\right) - 2\sqrt{7}\sin\left(\frac{\sqrt{7}}{2}\times 7\right)\right) + 14$

$= \mathrm{e}^{-\frac{7}{2}}\left(-10\cos\left(\frac{7\sqrt{7}}{2}\right) - 2\sqrt{7}\sin\left(\frac{7\sqrt{7}}{2}\right)\right) + 14$

$= 14.2716\ldots$

$b|_{t=7} = -8\mathrm{e}^{-\frac{1}{2}\times 7}\cos\left(\frac{\sqrt{7}}{2}\times 7\right) + 9$

$= -8\mathrm{e}^{-\frac{7}{2}}\cos\left(\frac{7\sqrt{7}}{2}\right) + 9$

$= 9.2383\ldots$

So when $t = 7$, the masses are slightly further away from their equilibrium than they were at $t = 6$

25) g)

The constant terms are the $+6$ and $+10$ in the original differential equations:

$\frac{\mathrm{d}a}{\mathrm{d}t} = -3a + 4b + 6$

$\frac{\mathrm{d}b}{\mathrm{d}t} = -2a + 2b + 10$

At the steady-state, the masses stop changing so $\frac{\mathrm{d}a}{\mathrm{d}t} = 0$ and $\frac{\mathrm{d}b}{\mathrm{d}t} = 0$:

$0 = -3a + 4b + 6 \Rightarrow 3a - 4b = 6$ (1)

$0 = -2a + 2b + 10 \Rightarrow 2a - 2b = 10$ (2)

Solve (1) & (2)

$a = 14$ and $b = 9$

So the constant terms shift the equilibrium away from $(0, 0)$

So the other coefficients determine how the solution approaches that steady state, while the constants determine **where** the steady state is.

26) a)

$\frac{\mathrm{d}p}{\mathrm{d}t} = 2p - 2r - 341\mathrm{e}^{-t}$

$\frac{\mathrm{d}r}{\mathrm{d}t} = 4p - 4r$

Rearrange the first equation:

$\frac{\mathrm{d}p}{\mathrm{d}t} = 2p - 2r - 341\mathrm{e}^{-t}$

$\Rightarrow r = -\frac{1}{2}\frac{\mathrm{d}p}{\mathrm{d}t} + p - \frac{341}{2}\mathrm{e}^{-t}$

$\frac{\mathrm{d}}{\mathrm{d}t}(r) = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{1}{2}\frac{\mathrm{d}p}{\mathrm{d}t} + p - \frac{341}{2}\mathrm{e}^{-t}\right)$

$\Rightarrow \frac{\mathrm{d}r}{\mathrm{d}t} = -\frac{1}{2}\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + \frac{\mathrm{d}p}{\mathrm{d}t} + \frac{341}{2}\mathrm{e}^{-t}$

Hence:

$\frac{\mathrm{d}r}{\mathrm{d}t} = 4p - 4r$

$\Rightarrow -\frac{1}{2}\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + \frac{\mathrm{d}p}{\mathrm{d}t} + \frac{341}{2}\mathrm{e}^{-t} = 4p - 4\left(-\frac{1}{2}\frac{\mathrm{d}p}{\mathrm{d}t} + p - \frac{341}{2}\mathrm{e}^{-t}\right)$

$\Rightarrow -\frac{1}{2}\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + \frac{\mathrm{d}p}{\mathrm{d}t} + \frac{341}{2}\mathrm{e}^{-t} = 4p + 2\frac{\mathrm{d}p}{\mathrm{d}t} - 4p + 682\mathrm{e}^{-t}$

$\Rightarrow -\frac{1}{2}\frac{\mathrm{d}^2p}{\mathrm{d}t^2} - \frac{\mathrm{d}p}{\mathrm{d}t} = \frac{1023}{2}\mathrm{e}^{-t}$

$\Rightarrow \frac{\mathrm{d}^2p}{\mathrm{d}t^2} + 2\frac{\mathrm{d}p}{\mathrm{d}t} = -1023\mathrm{e}^{-t}$

26) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + 2\frac{\mathrm{d}p}{\mathrm{d}t} = 0$

Auxiliary Equation:

$m^2 + 2m = 0$

$\Rightarrow m(m+2) = 0$

$\Rightarrow m = 0$ or $m = -2$

Complementary Function:

$p = A + B\mathrm{e}^{-2t}$

Try this Particular Integral:

$p = k\mathrm{e}^{-t}$

$\frac{\mathrm{d}p}{\mathrm{d}t} = -k\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2p}{\mathrm{d}t^2} = k\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + 2\frac{\mathrm{d}p}{\mathrm{d}t} = -1023\mathrm{e}^{-t}$

$\Rightarrow (k\mathrm{e}^{-t}) + 2(-k\mathrm{e}^{-t}) = -1023\mathrm{e}^{-t}$

$\Rightarrow k\mathrm{e}^{-t} - 2k\mathrm{e}^{-t} = -1023\mathrm{e}^{-t}$

$\Rightarrow -k\mathrm{e}^{-t} = -1023\mathrm{e}^{-t}$

$\Rightarrow -k = -1023$

$\Rightarrow k = 1023$

Particular Integral:

$p = 1023\mathrm{e}^{-t}$

General Solution:

$p = A + B\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t}$

26) c)

$p = A + B\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t}$

$\Rightarrow \frac{\mathrm{d}p}{\mathrm{d}t} = -2B\mathrm{e}^{-2t} - 1023\mathrm{e}^{-t}$

Then:

$r = -\frac{1}{2}\frac{\mathrm{d}p}{\mathrm{d}t} + p - \frac{341}{2}\mathrm{e}^{-t}$

$\Rightarrow r = -\frac{1}{2}(-2B\mathrm{e}^{-2t} - 1023\mathrm{e}^{-t}) + (A + B\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t}) - \frac{341}{2}\mathrm{e}^{-t}$

$\Rightarrow r = B\mathrm{e}^{-2t} + \frac{1023}{2}\mathrm{e}^{-t} + A + B\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t} - \frac{341}{2}\mathrm{e}^{-t}$

$\Rightarrow r = A + 2B\mathrm{e}^{-2t} + 1364\mathrm{e}^{-t}$

26) d)

$p = A + B\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t}$

Substitute $t = 0$ and $p = 2046$:

$2046 = A + B\mathrm{e}^{0} + 1023\mathrm{e}^{0}$

$\Rightarrow A + B = 1023$ (1)

$r = A + 2B\mathrm{e}^{-2t} + 1364\mathrm{e}^{-t}$

Substitute $t = 0$ and $r = 3411$:

$3411 = A + 2B\mathrm{e}^{0} + 1364\mathrm{e}^{0}$

$\Rightarrow A + 2B = 2047$ (2)

Solve (1) & (2):

$A = -1$ and $B = 1024$

$\therefore p = -1 + 1024\mathrm{e}^{-2t} + 1023\mathrm{e}^{-t}$ and $r = -1 + 2048\mathrm{e}^{-2t} + 1364\mathrm{e}^{-t}$

26) e)

Substitute $p = 0$ and $t = T$

$\Rightarrow 0 = -1 + 1024\mathrm{e}^{-2T} + 1023\mathrm{e}^{-T}$

$\Rightarrow 0 = -\mathrm{e}^{2T} + 1024 + 1023\mathrm{e}^{T}$

$\Rightarrow \mathrm{e}^{2T} - 1023\mathrm{e}^{T} - 1024 = 0$

$\Rightarrow (\mathrm{e}^{T} + 1)(\mathrm{e}^{T} - 1024) = 0$

$\Rightarrow \mathrm{e}^{T} \neq -1$ so $\mathrm{e}^{T} = 1024$

$\Rightarrow T = \ln 1024 = 6.931$ (3 DP)

26) f)

As $t \to \infty$, $\mathrm{e}^{-2t} \to 0$ and $\mathrm{e}^{-t} \to 0$

So $r \to -1$

The models predicts that the red squirrel population approaches -1, which is not realistic as a population cannot be negative. This means that the model is only reasonable up to a certain time and then breaks down.

27) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = a(x - y)$

$\frac{\mathrm{d}y}{\mathrm{d}t} = ay - bx$

Rearrange the first equation:

$\frac{\mathrm{d}x}{\mathrm{d}t} = ax - ay$

$\Rightarrow y = -\frac{1}{a}\frac{\mathrm{d}x}{\mathrm{d}t} + x$

$\frac{\mathrm{d}}{\mathrm{d}t}(y) = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{1}{a}\frac{\mathrm{d}x}{\mathrm{d}t} + x\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{a}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t}$

Hence:

$\frac{\mathrm{d}y}{\mathrm{d}t} = ay - bx$

$\Rightarrow -\frac{1}{a}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t} = a\left(-\frac{1}{a}\frac{\mathrm{d}x}{\mathrm{d}t} + x\right) - bx$

$\Rightarrow -\frac{1}{a}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\mathrm{d}x}{\mathrm{d}t} + ax - bx$

$\Rightarrow -\frac{1}{a}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + (b-a)x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2a\frac{\mathrm{d}x}{\mathrm{d}t} + a(a-b)x = 0$

27) b)

Auxiliary Equation:

$m^2 - 2am + a(a-b) = 0$

$\Rightarrow (m-a)^2 - a^2 + a(a-b) = 0$

$\Rightarrow (m-a)^2 = a^2 - a(a-b)$

$\Rightarrow (m-a)^2 = a^2 - a^2 + ab$

$\Rightarrow (m-a)^2 = ab$

$\Rightarrow m = a \pm \sqrt{ab}$

General Solution:

$x = A\mathrm{e}^{(a+\sqrt{ab})t} + B\mathrm{e}^{(a-\sqrt{ab})t}$

27) c)

$x = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$

$\Rightarrow \frac{dx}{dt} = A\left(a+\sqrt{ab}\right)e^{(a+\sqrt{ab})t} + B\left(a-\sqrt{ab}\right)e^{(a-\sqrt{ab})t}$

Then:

$y = -\frac{1}{a}\frac{dx}{dt} + x$

$\Rightarrow y = -\frac{1}{a}\left(A\left(a+\sqrt{ab}\right)e^{(a+\sqrt{ab})t} + B\left(a-\sqrt{ab}\right)e^{(a-\sqrt{ab})t}\right) + \left(Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}\right)$

$\Rightarrow y = -A\left(\frac{a+\sqrt{ab}}{a}\right)e^{(a+\sqrt{ab})t} - B\left(\frac{a-\sqrt{ab}}{a}\right)e^{(a-\sqrt{ab})t} + Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$

$\Rightarrow y = A\left(1-\frac{a+\sqrt{ab}}{a}\right)e^{(a+\sqrt{ab})t} + B\left(1-\frac{a-\sqrt{ab}}{a}\right)e^{(a-\sqrt{ab})t}$

$\Rightarrow y = A\left(\frac{a-(a+\sqrt{ab})}{a}\right)e^{(a+\sqrt{ab})t} + B\left(\frac{a-(a-\sqrt{ab})}{a}\right)e^{(a-\sqrt{ab})t}$

$\Rightarrow y = A\left(\frac{-\sqrt{ab}}{a}\right)e^{(a+\sqrt{ab})t} + B\left(\frac{\sqrt{ab}}{a}\right)e^{(a-\sqrt{ab})t}$

$\Rightarrow y = -A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}$

27) d)

$x(t) + \sqrt{\frac{a}{b}}y(t)$

$= Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t} + \sqrt{\frac{a}{b}}\left(-A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}\right)$

$= Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t} - Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$

$= 2Be^{(a-\sqrt{ab})t}$

$\therefore K_1 = 2B$

$x(t) - \sqrt{\frac{a}{b}}y(t)$

$= Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t} - \sqrt{\frac{a}{b}}\left(-A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}\right)$

$= Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t} + Ae^{(a+\sqrt{ab})t} - Be^{(a-\sqrt{ab})t}$

$= 2Ae^{(a+\sqrt{ab})t}$

$\therefore K_2 = 2A$

27) e)

$a+\sqrt{ab} > a-\sqrt{ab}$ for all $a, b > 0$

So the term $e^{(a+\sqrt{ab})t}$ dominates as $t \to \infty$

As $K_2 \neq 0$, we also have that $A \neq 0$

So for $x = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$, as $t \to \infty$ we have that $x \sim Ae^{(a+\sqrt{ab})t}$

And for $y = -A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}$, as $t \to \infty$ we have that

$y \sim -A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t}$

$\therefore$ as $t \to \infty$, we have that:

$$\frac{y(t)}{x(t)} \sim \frac{-A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t}}{Ae^{(a+\sqrt{ab})t}} = -\sqrt{\frac{b}{a}}$$

Hence:

$$\lim_{t\to\infty} \frac{y(t)}{x(t)} = -\sqrt{\frac{b}{a}}$$

27) f)

We have seen that from part e) that if $A \neq 0$ then $\lim_{t\to\infty} \frac{y(t)}{x(t)} = -\sqrt{\frac{b}{a}}$

However, if $A = 0$ then the term $e^{(a-\sqrt{ab})t}$ dominates as $t \to \infty$

So for $x = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$, if $A = 0$ and as $t \to \infty$ we have that $x \sim Be^{(a-\sqrt{ab})t}$

And for $y = -A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}$, if $A = 0$ and as $t \to \infty$ we have that

$y \sim B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}$

$\therefore$ as $t \to \infty$, we have that:

$$\frac{y(t)}{x(t)} \sim \frac{B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}}{Be^{(a-\sqrt{ab})t}} = \sqrt{\frac{b}{a}}$$

Hence:

$$\lim_{t\to\infty} \frac{y(t)}{x(t)} = \sqrt{\frac{b}{a}}$$

So the scientist's claim is incorrect.

28) a)

A decreases at a rate proportional to its amount, so:

$\frac{\mathrm{d}a}{\mathrm{d}t} = -pa$

where p is a positive constant.

B is gained from A, so there is a positive term proportional to a. This will be at the same rate as A decreases. B is lost as it is converted into C, so there is a negative term proportional to b. So:

$\frac{\mathrm{d}b}{\mathrm{d}t} = pa - qb$

where q is a positive constant.

28) b)

As no material is added or lost during the process, the total amount $a + b + c$ remains constant.

$\therefore \frac{\mathrm{d}}{\mathrm{d}t}(a + b + c) = 0$

$\Rightarrow \frac{\mathrm{d}a}{\mathrm{d}t} + \frac{\mathrm{d}b}{\mathrm{d}t} + \frac{\mathrm{d}c}{\mathrm{d}t} = 0$

Hence $\frac{\mathrm{d}c}{\mathrm{d}t} = -\frac{\mathrm{d}a}{\mathrm{d}t} - \frac{\mathrm{d}b}{\mathrm{d}t}$

So once $\frac{\mathrm{d}a}{\mathrm{d}t}$ and $\frac{\mathrm{d}b}{\mathrm{d}t}$ are known, $\frac{\mathrm{d}c}{\mathrm{d}t}$ is already determined. Hence the third differential equation is not independent.

29) a)

Use Separation of Variables to solve $\frac{\mathrm{d}p}{\mathrm{d}t} = -2p$

$\Rightarrow \int \frac{1}{p}\mathrm{d}p = \int -2\,\mathrm{d}t$

$\Rightarrow \ln|p| = -2t + c$

$\Rightarrow p = A\mathrm{e}^{-2t}$

Substitute $t = 0$ and $p = 10$:

$10 = A\mathrm{e}^{0}$

$\Rightarrow A = 10$

$\therefore p = 10\mathrm{e}^{-2t}$

29) d)

Since no material is lost during the process, the total amount of material remains constant.

$p + q + r = k$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(p + q + r) = 0$

$\Rightarrow \frac{\mathrm{d}p}{\mathrm{d}t} + \frac{\mathrm{d}q}{\mathrm{d}t} + \frac{\mathrm{d}r}{\mathrm{d}t} = 0$

29) c)

This means that the total amount of material in the system does not change with time. As P is converted into Q, and Q is converted into R, material is only transferred between the three substances. None is created or lost.

29) d)

$\frac{\mathrm{d}q}{\mathrm{d}t} = 2p - q$

$\Rightarrow \frac{\mathrm{d}q}{\mathrm{d}t} = 2(10\mathrm{e}^{-2t}) - q$

$\Rightarrow \frac{\mathrm{d}q}{\mathrm{d}t} = 20\mathrm{e}^{-2t} - q$

$\Rightarrow \frac{\mathrm{d}q}{\mathrm{d}t} + q = 20\mathrm{e}^{-2t}$

Integrating Factor

$= \mathrm{e}^{\int 1\mathrm{d}t}$

$= \mathrm{e}^{t}$

Let Integrating Factor $= \mathrm{e}^{t}$

$\frac{\mathrm{d}q}{\mathrm{d}t} + q = 20\mathrm{e}^{-2t}$

$\Rightarrow \mathrm{e}^{t}\frac{\mathrm{d}q}{\mathrm{d}t} + \mathrm{e}^{t}q = 20\mathrm{e}^{-2t} \times \mathrm{e}^{t}$

$\Rightarrow \mathrm{e}^{t}\frac{\mathrm{d}q}{\mathrm{d}t} + \mathrm{e}^{t}q = 20\mathrm{e}^{-t}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{t}q) = 20\mathrm{e}^{-t}$

$\Rightarrow \mathrm{e}^{t}q = -20\mathrm{e}^{-t} + c$

$\Rightarrow q = -20\mathrm{e}^{-2t} + c\mathrm{e}^{-t}$

Substitute $t = 0$ and $q = 0$:

$0 = -20\mathrm{e}^{0} + c\mathrm{e}^{0}$

$\Rightarrow 0 = -20 + c$

$\Rightarrow c = 20$

$\therefore q = -20\mathrm{e}^{-2t} + 20\mathrm{e}^{-t}$

29) e)

We know that $p + q + r = 10 + 0 + 0 = 10$

$\therefore (10\mathrm{e}^{-2t}) + (-20\mathrm{e}^{-2t} + 20\mathrm{e}^{-t}) + r = 10$

$\Rightarrow 10\mathrm{e}^{-2t} - 20\mathrm{e}^{-2t} + 20\mathrm{e}^{-t} + r = 10$

$\Rightarrow r = 10 + 10\mathrm{e}^{-2t} - 20\mathrm{e}^{-t}$

30) a)

Use Separation of Variables to solve $\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{3}{100}x$

$\Rightarrow \int \frac{1}{x}\mathrm{d}x = \int -\frac{3}{100}\mathrm{d}t$

$\Rightarrow \ln|x| = -\frac{3}{100}t + c$

$\Rightarrow x = A\mathrm{e}^{-\frac{3}{100}t}$

Substitute $t = 0$ and $x = 100$:

$100 = A\mathrm{e}^{0}$

$\Rightarrow A = 100$

$\therefore x = 100\mathrm{e}^{-\frac{3}{100}t}$

30) b)

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{100}x - \frac{1}{100}y$

$\Rightarrow x = \frac{100}{3}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{3}y$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{3}{100}x - \frac{1}{100}y\right)$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{3}{100}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{1}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{3}{100}\left(-\frac{3}{100}x\right) - \frac{1}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\frac{9}{10000}x - \frac{1}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\frac{9}{10000}\left(\frac{100}{3}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{3}y\right) - \frac{1}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\frac{3}{100}\frac{\mathrm{d}y}{\mathrm{d}t} - \frac{3}{10000}y - \frac{1}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{1}{25}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{3}{10000}y = 0$

30) c)

Auxiliary Equation:

$m^2 + \frac{1}{25}m + \frac{3}{10000} = 0$

$\Rightarrow \left(m + \frac{1}{100}\right)\left(m + \frac{3}{100}\right) = 0$

$\Rightarrow m = -\frac{1}{100}$ or $m = -\frac{3}{100}$

General Solution:

$y = B\mathrm{e}^{-\frac{1}{100}t} + C\mathrm{e}^{-\frac{3}{100}t}$

Substitute $t = 0$ and $y = 0$:

$0 = B\mathrm{e}^0 + C\mathrm{e}^0$

$\Rightarrow B + C = 0$ (1)

To get a second condition, consider the original DE:

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{100}x - \frac{1}{100}y$

At $t = 0$, we have $x = 100$ and $y = 0$, so:

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3}{100} \times 100 - \frac{1}{100} \times 0 = 3$

Differentiate the general solution:

$y = B\mathrm{e}^{-\frac{1}{100}t} + C\mathrm{e}^{-\frac{3}{100}t}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{100}B\mathrm{e}^{-\frac{1}{100}t} - \frac{3}{100}C\mathrm{e}^{-\frac{3}{100}t}$

Substitute $t = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 3$:

$3 = -\frac{1}{100}B\mathrm{e}^0 - \frac{3}{100}C\mathrm{e}^0$

$\Rightarrow 3 = -\frac{1}{100}B - \frac{3}{100}C$

$\Rightarrow 300 = -B - 3C$

$\Rightarrow B + 3C = -300$ (2)

Solve (1) & (2):

$B = 150$ and $C = -150$

$\therefore y = 150\mathrm{e}^{-\frac{1}{100}t} - 150\mathrm{e}^{-\frac{3}{100}t}$

30) d)

As the total amount of material remains constant, at any time we have:

$x + y + z = 100$

$\Rightarrow z = 100 - x - y$

$\Rightarrow z = 100 - \left(100\mathrm{e}^{-\frac{3}{100}t}\right) - \left(150\mathrm{e}^{-\frac{1}{100}t} - 150\mathrm{e}^{-\frac{3}{100}t}\right)$

$\Rightarrow z = 100 - 100\mathrm{e}^{-\frac{3}{100}t} - 150\mathrm{e}^{-\frac{1}{100}t} + 150\mathrm{e}^{-\frac{3}{100}t}$

$\Rightarrow z = 100 - 150\mathrm{e}^{-\frac{1}{100}t} + 50\mathrm{e}^{-\frac{3}{100}t}$

As $t \to \infty$, $\mathrm{e}^{-\frac{1}{100}t} \to 0$ and $\mathrm{e}^{-\frac{3}{100}t} \to 0$

So $z \to 100$ kg